

SLOOP 2.8.1 — The Complete Guide

Everything SLOOP does, every button, every combination, every page, in one place. This guide is written to be read from the top the first time, then used as a reference: the cheat sheet at the end has every combination on one page.

SLOOP is a free, open-source (GPL-3.0) firmware for the M-VAVE FM-1, based on Felucca by Leo Kuroshita / Hügelton Instruments. The short manual is SLOOP.md; the project page is README.md.

How to read this guide

Written	Means
tap FX	press FX and let go without touching anything else (shorter than about half a second)
hold FX	keep FX pressed
FX + key 3	hold FX, and while it is held press white key 3
FX + KNOB 2	hold FX, and while it is held turn KNOB 2
SEQ + step + KNOB 4	hold SEQ, hold a step key, turn KNOB 4
key 1 … key 16	the 16 white keys, from the lowest (**F3** = key 1) to the highest (**G5** = key 16)
GO	a value that does something (LOAD, SAVE, NEW…): turn it one click, *AGAIN:* … shows, turn it one more click within about 1.5 s to do it

The 16 white keys, numbered:

Key	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
Note	F3	G3	A3	B3	C4	D4	E4	F4	G4	A4	B4	C5	D5	E5	F5	G5

The 11 black keys are F#3, G#3, A#3, C#4, D#4, F#4, G#4, A#4, C#5, D#5, F#5. The "first four black keys" are F#3, G#3, A#3 and C#4.

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What's new in 2.8

3,000 of you now follow SLOOP — thank you. 2.8 is a deeper way to write and shape your music, and most of it comes from your comments and reports. Each item points to its full write-up below.

- **A NOTES editor: a piano roll for the synth tracks.** Tap **SEQ** on a synth track for the pattern as a piano roll, and write and edit notes step by step: KNOB 1 the notes (a chord's one by one), KNOB 2 the pitch, KNOB 3 the length, KNOB 4 the force; PRESETS zooms, EDIT deletes. Hold **OCT+** on a step for its **sheet** — SLIDE, NUDGE, FILL, MOVE, COPY, DELETE. See The step sequencer.
- **The same per-step sheet on the DRUMS grid.** Hold **OCT+** on a drum step for RATCHET, NUDGE, FILL, MOVE, COPY, DELETE — the groove, shaped step by step. See The step sequencer.
- **The playing section on the main screen.** The TRACKS screen shows which section — **A to D** — is playing, in its own colour. See The screens.
- **Filter resonance.** The per-track filter gains a **resonance**, 0 (clean, as before) to 10: HOME menu → AUDIO → **FILTER RESO**. See Per track.
- **The FX filter on the drum track too.** FX → FILTER now works on drums: sweep and shape the whole kit, live. See Per track.
- **Drum TUNE on the noise sounds.** On the DRUMS Sound page, **TUNE** now also shapes the brightness of the tone-less sounds (closed hats, shakers, claps). See The drum track.
- **A smoother punch-in.** A settable fade-in and out for the 27 punch-in effects: hold **FX**, then **SELECT**, 0 (instant) to 15 (~0.37 s). See Punch-in effects.
- **Steadier external MIDI clock.** The incoming tempo is read from the average of the last beats, so a steady 140 no longer wavers 139 / 140 / 141; a real tempo change is taken at once. See MIDI out, clock in.
- **MIDI Bank Select (CC 0).** Reach every preset past the 128 Program Change range — the rest of the factory sounds, then your own user presets by slot. See MIDI CCs.
- **USB RESCUE, fixed.** Holding **OCT–** at power-on now reliably opens USB rescue mode. See Rescue and special modes.
- **Going back to stock changed.** The in-browser return-to-stock was removed (it was not reliable on every system); use M-VAVE's own updater **M-UPGRADE**. See Going back.

Everything from 2.7 is unchanged and still here. Projects, autosaves and backups from 2.7 load as they are, and a project saved in 2.8 opens in 2.7 too.

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1. Install and update

From the browser (recommended). Open the installer page in **Chrome or Edge**, connect the FM-1 by USB (a data cable, plugged straight in, no hub), press **INSTALL**, allow MIDI access, and wait for *Done*. The FM-1 restarts on the SLOOP logo.

From the source folder (Windows). Double-click `INSTALL-SLOOP.bat`: it builds the firmware and opens the installer at `http://localhost:8766/webapp/installer/`. Keep the black window open until the install is done. `OPEN-EDITOR.bat` opens the web editor.

Updating keeps your projects, presets, samples, kits and settings. The installer checks the package (SHA-256), SLOOP's update loader checks it again (CRC) before it starts the new firmware, and a damaged package is refused before anything is written.

Before you go back to an older SLOOP (2.3 or earlier), save a backup in the editor: 2.3 cannot read 2.4 projects (format FUN5).

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2. The panel

Control	Kind	Main job
PLAY	button	start / stop all four tracks
REC	button	record (tap), clear the track (hold)
SAVE	button	song layer (hold), song screen / save pages (tap)
HOME	button	TRACKS screen, visualiser (tap), menu (hold), lock a layer
ENV	button	envelope pages
LFO	button	LFO pages
FX	button	punch-in layer (hold), effect pages (tap)
SEL	button	key / chord layer (hold), scale pages (tap). The second button of the top row, between FX and ENV (not the SELECT knob)
EDIT	button	erase layer (hold), engine pages (tap)
ARP	button	note-repeat layer (hold), arpeggiator pages (tap)
SEQ	button	step layer (hold), sequencer pages (tap)
GLO	button	mix layer (hold), global pages (tap)
OCT– / OCT+	buttons	octave, ghost / hard hits, undo / redo, pages
ALGORITHM	knob	selects the track (1–4)
PRESETS	knob	the selected track's sound, or the drum kit
SELECT	knob	tempo, or the previous / next page
KNOB 1–4	knobs	what the four dials at the bottom of the screen show
MASTER	knob	output volume (headphones / line out)
27 keys	keys	16 white (F3–G5) and 11 black

3. The big ideas: tracks, colours, tap and hold

Four tracks

Track	Colour	Plays	Knob of the same colour
1	blue	a synth (3 synth tracks share 8 voices)	KNOB 1
2	green	a synth	KNOB 2
3	yellow	a synth	KNOB 3
4	orange	the drum machine: 16 sounds, one per white key	KNOB 4

ALGORITHM picks the track you play, record and edit, on every screen. The screen and the whole editor take the colour of the selected track. **White** on the screen always means *what you are touching*; **red** always means *recording*.

Tap and hold

Every function button has two lives:

- **Tap** it: its **pages** open (sound design, settings), as on any FM-1 firmware. Tap it again: the next page of that button. **SELECT** also walks the pages of that button, both ways. Each button remembers the last page you used.
- **Hold** it: a **layer**. While it is held, the 16 white keys and KNOB 1–4 do something else, and after 0.14 s the screen shows the 16 keys as 16 tiles (four rows of four: keys 1–4, 5–8, 9–12, 13–16) and the knobs as dials. Let go: back to playing.

The layers are **FX** (punch-in effects), **EDIT** (erase), **ARP** (note repeat), **SEQ** (steps), **SEL** (key and chords), **GLO** (mix) and **SAVE** (song). ENV and LFO have no layer: they only open pages.

Landmarks: while a layer is held (and always on the drum track) keys 1, 5, 9 and 13 glow dimly — the first key of each row of tiles — so you can find a tile without looking. Keys at full light are what is on.

Lock a layer: hold its button and **tap HOME**. The layer stays open with the button let go (*LOCK* on the screen, the button blinks): both hands free, one on the keys, one on the knobs. Any other button lets it go (HOME, the layer's own button, ENV…) and does only that; **PLAY, REC and OCT– / OCT+ keep working** inside a locked layer.

Inside any layer: PLAY still starts and stops, **SELECT** is always the tempo, and the ALGORITHM / PRESETS knobs wait (no jump when you let go).

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4. Your first beat in 60 seconds

1. Turn **ALGORITHM** to track **4** (orange). The white keys play 16 drum sounds: **F3 kick**, G3 kick 2, A3 snare, B3 clap, **C4 hat**, D4 open hat… Turn **PRESETS** to pick a kit: try *808* or *BOOMBAP*.
2. Press **REC** (*rec ready*). **Play a beat freely**, at your own tempo — no click, no count-in. Hold **OCT–** while you hit for ghost notes, **OCT+** for hard hits.
3. **Press REC on the "1" after your last bar.** The loop closes: its length sets the tempo, the hits snap to the grid, and it plays at once.
4. Press **REC** again while it plays: you record on top (overdub). Hold **ARP** and hold the hat key (C4): a 1/16 hat roll, recorded as ratchets.
5. Turn **ALGORITHM** to track **1** (blue, *808 BOOM*), press **REC**, play a bass line.
6. Hold **SEL** and press the key of your song (for example D). Select track 2, hold **SEL** and turn **KNOB 1** to *7TH*: every white key is now a chord of the key, one finger each.
7. Hold **FX** and hold a key (white or black) for a punch-in effect; still holding FX, turn **KNOB 2** for DUST, **KNOB 3** for DUCK.
8. A mistake? **EDIT + OCT–**: undo.

5. Every button, alone

Control	Tap / turn	Hold
PLAY	start / stop all tracks (in song mode: play the song). Works inside every layer	—
REC	playing: record now / stop recording · stopped: arm (*rec ready*) · free take: close the loop · count-in: cancel	~2 s: **clear the selected track** (a ring fills; let go before it is full and nothing happens)
SAVE	on TRACKS or DRUMS: the **SONG** screen · on the SONG screen: save the song · elsewhere: the **SAVE pages**	the **song layer** (sections A–D, chain, song mode, SONG REC)
HOME	the **TRACKS** screen · on TRACKS: the **visualiser** (HOME again closes it)	0.7 s: the **menu** (SCREEN, LIGHTS, AUDIO, SYSTEM); hold again to leave it
ENV	ENV pages (ENV, ENV DEST)	—
LFO	LFO pages (LFO, LFO DEST)	—
FX	FX pages (FX, FILTER, SLICER, DLY, REV/CHO)	punch-in layer
SEL	SEL pages (SEL, SEL 2)	key / chord layer
EDIT	EDIT pages (EDIT 1, EDIT 2, VOICE, VOICE 2) · on TRACKS with the drum track: the **DRUMS** screen · on DRUMS: grid ↔ kit · on the NOTES page: delete the note	erase layer
ARP	ARP pages (ARP, ARP 2)	note-repeat layer
SEQ	SEQ pages (STEP, PATTERN, SONG) · with the drum track: the **DRUMS** screen (2.7.1: from any page; it opened STEP from ARP's) · on DRUMS: grid ↔ kit	step layer
GLO	GLO pages (GLOBAL, MASTER, SYSTEM, DRUMS)	mix layer
OCT– / OCT+	synth track: octave down / up (–3 … +3) · both together: octave 0	drum track: **ghost** (OCT–) / **hard** (OCT+) hits while held
ALGORITHM	select track 1–4, on every screen (inside a layer it waits; not during a free take, on the SONG screen or in the menu)	—
PRESETS	on TRACKS, HOME and the PRESETS page: the selected track's **sound** (your favourites first (2.7), then all factory sounds by kind, then your user presets) · drum track: the **kit** · on the DRUMS screen: the kit	—
SELECT	on TRACKS, inside a layer and on the REC screen: the **tempo** · on a page: previous / next page of its button (stops at the ends) · DRUMS screen: grid ↔ kit · visualiser: the style · menu: the section	—
KNOB 1–4	the four dials at the bottom of the screen	—
MASTER	output volume (the visualiser ignores it)	—

Combinations at power-on and while stopped

Combination	Does
OCT– held while switching on	**USB rescue** (*SLOOP USB RESCUE*): install again from the browser
OCT– + OCT+ held while switching on	**hardware calibration**: teach each button and knob (also in the HOME menu → SYSTEM)

Combination	Does
OCT– + OCT+ held 5 s, stopped	**update mode** (UBOOT): a countdown shows from 2 s; let go to cancel. Your work is saved first

6. Playing the keys

Synth tracks (1–3)

- The keys play the selected track. The notes you play always go out on USB MIDI (the track's channel).
- **OCT– / OCT+**: octave –3 … +3; both together: back to 0. The OCT lights show when you are off 0.
- What the keys play depends on the track's **KEYS** setting and **CHORD** mode (SEL layer or SEL page):

KEYS	CHORD	The keys play
OFF	OFF	every key chromatic (the default)
SNAP	OFF	every key, rounded down to the scale
WHITE	OFF	the white keys walk the scale from C4 (C4 = the key's root); black keys are silent
any	**on**	the white keys play **chords** of the scale, one per key (C4 = the I chord); the black keys **change the chord** (CHORD+)

- The **GM KIT** sound (SAMPLE engine) is the exception: the lowest key is a kick (C2), no scale.

Drum track (4)

16 sounds, one per white key; a black key plays the sound of the white key on its left (two fingers on one sound, for fast rolls). Hold **OCT–** for ghost notes, **OCT+** for hard hits. See The drum track.

Velocity and levels

The FM-1's keys have no velocity: on the drum track every hit has one of four **levels** — **GHOST**, **SOFT**, **NORM** (as played), **HARD** — chosen with OCT– / OCT+ as you play, or later in the step sequencer. On synth tracks a note has a **FORCE** too, and a step an **accent** (NOTES page, KNOB 4).

7. The drum track

The 16 sounds

Key	Sound	Key	Sound	Key	Sound	Key	Sound
1 · F3	KICK	5 · C4	HAT	9 · G4	SNARE 2	13 · D5	RIDE
2 · G3	KICK 2	6 · D4	OPEN HAT	10 · A4	LOW TOM	14 · E5	SHAKER
3 · A3	SNARE	7 · E4	PEDAL HAT	11 · B4	HI TOM	15 · F5	CONGA
4 · B3	CLAP	8 · F4	RIM	12 · C5	CRASH	16 · G5	COWBELL

- **Levels:** GHOST, SOFT, NORM, HARD (hold OCT– / OCT+ while you hit). Softer hits are darker as well as quieter.
- **Ratchets:** a hit can repeat x1–x4 inside its step (ARP rolls record them; SEQ + step + KNOB 3 sets them).
- **Choke:** the closed hat and the pedal hat cut the open hat.
- **6 drum voices**, separate from the synths' 8.

Kits

PRESETS on the drum track (or KNOB 1 on the DRUMS kit page, or the editor) picks the kit:

#	Kits
1–5	**ACOUSTIC** (sampled studio kit, CC0) and its treatments: DEEP, TIGHT, BRIGHT, DUST
6–37	32 synthesised kits, 16 sounds each: 808, 909, 606, 80S, VINTAGE, TRAP, DRILL, BOOMBAP, LO-FI, PHONK, HOUSE, D.HOUSE, TECHNO, MINIMAL, ELECTRO, DISCO, GARAGE, JUNGLE, DUBSTEP, DEMBOW, AMAPIANO, AFRO, LATIN, TRIBAL, SYNTHWV, CHIP, ARCADE, GLITCH, INDUSTR, HYPER, AMBIENT, JAZZ
38–41	**USR1, USR2, USR3, USR4**: your own kit in one sample slot (about 7.4 s)
42	**USR3+4**: your own big kit over two slots (about 15 s)
43–46	**SYN1, SYN2, SYN3, SYN4** (2.5): your own synthesised kits, made in the editor's Drum synth page. The FM-1 shows the name you gave them

Every kit is level-matched. The kit is saved with projects and song sections. Changing the kit never touches the pattern. Your own kits work like any other: steps, levels, ratchets, fills, choke, MIDI.

The DRUMS screen

Open it with **EDIT** or **SEQ** tapped on the TRACKS screen while the drum track is selected. It has three pages, grid, kit and (2.7) sound; **EDIT / SEQ tapped** goes to the next one, **SELECT** right / left walks them.

Grid page — the 16 sounds × 16 steps, levels as shades, ratchets as notches.

Control	Does
KNOB 1 **sound**	picks the sound (lane); (2.7.1) heard only with MENU → SEQ → **DRUM PREVIEW ON** (OFF by default, quiet for live)
KNOB 2 **step**	moves to a step (DRUM PREVIEW ON: you hear what it holds)
KNOB 3 **hit**	right: a hit of that sound on that step (NORM) · left: clear it

Control	Does
KNOB 4 **level**	the hit's level up / down (GHOST, SOFT, NORM, HARD)
white keys	the 16 steps of the sound picked: press to set the step, again to clear it (2.7.1: silent, the pattern plays it); the keys light that sound's steps
black keys (2.7.1)	pick the sound, silently: F#3 KICK / KICK 2 · G#3 SNARE / SNARE 2 · A#3 RIM · C#4 CLAP · D#4 HAT / PEDAL · F#4 OPEN HAT · G#4 LOW TOM / HI TOM · A#4 CRASH · C#5 RIDE · D#5 SHAKER / CONGA · F#5 COWBELL (the same key again: the other sound of its pair); the black key of the sound picked is lit
OCT– / OCT+ (2.7.1)	the page of steps: 1–16, 17–32, 33–48, 49–64 (round; up to the track's length). Up to 2.7: the first four black keys
PRESETS	the kit
SAVE	the SONG screen
HOME	back to TRACKS

Kit page — 16 pads that flash on every hit; the keys play the pads.

KNOB 1	KNOB 2	KNOB 3	KNOB 4
kit	**level** of the drum track	**reverb** send of the drums	**pan** of the drum track

Sound page (2.7) — each of the 16 sounds tuned and shortened or lengthened, on every kit. Play a key (or turn KNOB 1) to pick the sound: its pad is framed, and a sound you changed shows a small square.

KNOB 1	KNOB 2	KNOB 3	KNOB 4
sound (you hear it)	**tune**: –8 … +7 semitones	**decay**: –4 (very short) … 0 (as the kit has it) … +3 (long)	**level** of the drum track

On the synthesised kits (808, 909… and your SYN1–4) DECAY shortens or lengthens the sound itself; on the sampled kits (ACOUSTIC … DUST, USR) it can only shorten it (a fade), +3 plays the sample as it is. **(2.8)** On the synthesised kits, TUNE also shapes the **brightness** of the tone-less sounds (closed hats, shakers, claps), which have no pitch to shift: lower it for a darker, duller hat, raise it for a crisper one. The settings are kept with the project and with each section A–D, so a section can have its own tuning; changing the kit keeps them, NEW sets them back to 0. A project from an older SLOOP opens with every sound as its kit has it.

While the song plays in song mode, the grid's hit and level knobs wait (*STOP THE SONG FIRST*).

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8. Recording

SLOOP records live and layers every pass on top of the last (overdub). Notes go to the nearest step **as you heard it**: the ~12 ms between a key and its sound are taken back, so what you play on the beat lands on the beat. Chords are kept on the synth tracks whose VOICE is POLY (up to 4 notes a step; a MONO, LEGATO or UNISON sound, as most basses and leads, plays and records one note at a time); held notes become ties. (2.7.1) On a POLY track each held note goes on until its own key is let go, even while other notes start and end over it: hold a note, play others on top, and it keeps sounding in the pattern (up to 4 at once). (2.7.2) On a MONO, LEGATO or UNISON track, a note played while the one before is still held is recorded **legato** and plays back as you played it: with LEGATO the sound goes on into the new note without a new attack (it glides only with GLIDE or a SLIDE), where before the sequencer ended the note and started the next one again. It works from the FM-1's keys and from a MIDI controller alike. Let go of the second note while the first is still held: the pattern does not go back to the first one (the live sound does). The TRACKS page draws one mark a step; the NOTES page shows a chord's notes one by one (*2/3*…).

Over held notes (2.7). On a POLY track, a note you record while held notes of the pattern ring (a pad tied over the bar) no longer cuts them: they go on under it, and your note ends where you let it go. Up to 4 notes sound at once on a track (a fifth makes the oldest held one end there). Entering steps by hand (SEQ + key) still replaces what plays, so a long note can be cut on purpose. On SLOOP 2.6 such a pattern plays the held notes again where a note was laid over them.

With a MIDI keyboard: REC records the **selected** track. A keyboard on channel 1, 2 or 3 always plays track 1, 2 or 3 (2.6: the channels of GLO → MIDI), whatever track is selected: set it to channel 4 (or 4–16) to play and record the selected track. From 2.6, if your keyboard plays another track than the one recording, the screen says *MIDI PLAYS TRACK n* (once a take).

What REC does

When	REC does	Then
Playing	records the selected track **at once**	REC again stops recording; the loop plays on
Stopped, project with notes	arms (*rec ready*, the REC light blinks)	**your first note starts the loop and is step 1**; PLAY starts it too
Stopped, empty project	arms (*rec ready · play freely*)	a **free take**, or, with MODE *tempo*, as with notes

REC again while armed disarms it (*REC OFF*). In **song mode**, REC waits until the song is stopped (*STOP THE SONG FIRST*), and arming REC switches back to loop mode.

While recording, the REC light is solid and the track shows a red *rec*. **Turn ALGORITHM** while recording and the take moves to the next track without stopping.

The REC screen (armed, before the first note)

Knob	Dial	Choices
KNOB 1	**mode** (empty project only)	**free**: a free take, the tempo follows you · **tempo**: record at the tempo set (SELECT)
KNOB 2	**length**	the selected track's loop: **1, 2 or 4 bars**

Knob	Dial	Choices
KNOB 3	**start**	**note**: your first note starts the loop · **count**: press **PLAY**, one bar of clicks (4, 3, 2, 1 on the screen), then recording starts; notes before only sound
KNOB 4	**quant** (2.7)	**on**: what you record lands on the steps (as before) · **off**: it keeps your timing (see below)

LENGTH and START apply when you record at a tempo (MODE *tempo*, or a project that already has notes); a free take makes its own length. MODE, START and QUANT are settings of the FM-1 (they stay as you left them, also for recording while the song plays).

QUANT off (2.7). A note you play still goes into the nearest step, and how far off the grid you played it becomes that step's **nudge** (SEQ + step + KNOB 4: 1/64 of a step, about 2 ms at 120 BPM in 1/16). So the pattern plays back with your groove, and you can still edit it step by step (a nudged step shows its dot; SEQ + step + OCT– puts it back on the grid). It works for the keys and MIDI, in the free take too; the arp and note repeat stay on their grid. A step has one nudge: the first note in a step sets it, and the other notes of a chord or other drum sounds on that step follow it (in DIV 1/32 the steps are twice as fine). Undo (EDIT + OCT–) takes a take's nudges back with its notes. During the count-in, **REC cancels it** and **PLAY** goes back to *rec ready*. SELECT sets the tempo on this screen.

Free take: the loop follows you

1. On an empty project, **REC**, then play freely. The screen shows *free take*, the seconds, and the loop it would make right now (*2 bars · 92 bpm*).
2. **Press REC on the "1" after your last bar.** SLOOP picks 1, 2 or 4 bars at the tempo nearest the one set (within 3 % the set tempo is kept), writes your notes with their lengths and levels, and plays the loop at once. All four tracks take that length.
3. **PLAY** during a free take drops it. A take closes by itself after 24 s.

Metronome and swing

- The **PLAY light flashes on every beat**: a silent metronome.
- An audible click: **GLO** → **GLOBAL** → **CLICK** = **OFF**, **REC** (only while a track records) or **ON** (whenever it plays). It is a short tick of its own (higher on the first beat of a bar), heard with every kit, even with the drum track muted or another track soloed; its level follows GLO → DRUMS → LVL. The click is never recorded, and the count-in always clicks.
- **Swing** goes from **0** (straight) to **100** (the strongest: a step pair played 75 / 25, as MPC swing at 75 %): **HOME, KNOB 1** or **GLO** → **GLOBAL** → **SWING** for all tracks, **SEQ + KNOB 3** (or SEQ → PATTERN → SWG) per track; a track's swing adds to the global one. Triplet divisions are never swung.

What gets recorded

- **Synth tracks:** the notes (and chords) with their lengths; ties for held notes; ARP rolls as ratchets; CHORD+ chords as they sound.
- **Drum track:** the hits with their levels; ARP rolls as **ratchets** (a 1/32 roll on a 1/16 track: x2 on each step).
- Never recorded: punch-in effects, the click, notes played in a layer that uses the keys for something else (FX, SEQ, GLO…).

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9. The layers, one by one

All seven layers at a glance:

Hold	Name	White keys	KNOB 1	KNOB 2	KNOB 3	KNOB 4	Tap
FX	punch	a punch-in effect while the key is held	FILTER (master)	DUST	DUCK	TRK FILT (selected track)	FX pages
EDIT	erase	erase that sound / note from the pattern	SHIFT	LENGTH ×2 / ½	TRANSPOSE	—	EDIT pages
ARP	roll	note repeat on the grid	RATE	—	—	—	ARP pages
SEQ	steps	steps 1–16 of the page	SOUND / NOTE	DIV	SWING	LENGTH	SEQ pages
SEL	key	the key of the song	CHORD	SCALE	KEYS	TRANSPOSE	SEL pages
GLO	mix	1–4 mute · 5–8 solo · 9 fill · 10 fill bar · 16 tap tempo	level 1	level 2	level 3	level 4	GLO pages
SAVE	song	1–4 play A–D · 5–8 save A–D · 13 loop / song · 14 SONG REC · 16 song screen	—	—	—	—	song screen / SAVE pages

FX — punch

Control	Does
FX + a key	the punch-in effect of that key (16 white keys, and 11 black keys, 2.6), on the whole mix, for as long as the key is held
FX + KNOB 1	master **FILTER**: left a low-pass, right a high-pass, centre off
FX + KNOB 2	master **DUST** (old sampler and vinyl)
FX + KNOB 3	master **DUCK** (sidechain pump from the kick)
FX + KNOB 4	**TRK FILT**: the same one-knob filter on the **selected track only** (any track, the drums too)
FX + OCT– (held, 2.6)	**REV THROW**: every track's reverb send (the drums' too) goes full while held, gliding in and out (~12 ms); with a punch-in key too. The song's REV values never change, nothing is saved or recorded
FX + OCT+ (held, 2.6)	**ECHO THROW**: the same for the delay sends, and FDBK at least 70 % while held. Both together: both throws
FX + HOME	lock the FX layer open (both hands free)

Keys pressed while FX is held never play or record notes.

EDIT — erase

Control	Does
EDIT + key, playing	erases that sound (drums) or note (synths; with CHORD on, its chord) from **every step the playhead passes while you hold the key** — MPC style: hold the hat key for one bar and that bar's hats are gone. *ERASED* flashes
EDIT + key, stopped	erases that sound / note from the **whole pattern** at once
EDIT + KNOB 1	**SHIFT**: every step one later (right) / earlier (left) — turns the groove around

Control	Does
EDIT + KNOB 2	**LENGTH**: right ×2 (the pattern copied after itself, up to 64 steps), left ½
EDIT + KNOB 3	**TRANSPOSE**: every note a semitone up / down (synth tracks)
EDIT + OCT–	**undo**
EDIT + OCT+	**redo**
EDIT + a black key (drum track, 2.6)	**MUTE mode**: the white keys now **mute / unmute their sound** in the sequencer, live, without erasing anything (the muted sounds show *OFF* in red; the screen says *mute: key = sound*). A black key again goes back to erasing. 2.7: MUTE mode **stays** when you let go of EDIT, so the next EDIT + key mutes at once (2.6 went back to erasing)

All the knob turns of one EDIT hold count as one change for undo.

The drum mutes stay on until you unmute the sound (EDIT + a black key, then its key again); they are a performance tool, not part of the pattern: a project loaded or a new one starts with every sound playing. A muted sound still plays from the keys and from MIDI, and its ratchets are muted too.

ARP — roll (note repeat)

Control	Does
ARP + key (held)	the key repeats on the grid at the RATE, locked to the tempo and the swing; it ends with the key
ARP + KNOB 1	**RATE**: 1/8, 1/16, 1/32, 32T, 1/64 (also GLO → MASTER → ROLL)
ARP + key + OCT– / OCT+	on the drum track: ghost / hard rolls

While recording, a roll is written as ratchets (a 1/32 roll on a 1/16 track: x2 on each step). Rolls ignore fills. (The **arpeggiator** is something else: tap ARP for its pages.)

SEQ — steps

The 16 white keys are the 16 steps of the current page; the lit ones play.

Control	Does
SEQ + empty step	sets it at once — drums: with the sound shown (KNOB 1, or the last pad you hit); synths: with the note or chord you played last (on the keys, or a MIDI keyboard: 2.6)
SEQ + set step (tap)	clears it (with its nudge, locks and fill condition)
SEQ + several step keys held	edit them together
SEQ + first four black keys or **SEQ + OCT– / OCT+** (no step held)	the page: steps 1–16, 17–32, 33–48, 49–64 (up to the track's length). (2.7.1) The drum track: OCT– / OCT+ only, the black keys pick the sound
SEQ + a black key (drum track, 2.7.1)	the **sound** the steps are for, silently, as on the DRUMS grid (F#3 KICK / KICK 2 … F#5 COWBELL; the same key again: its pair). With **SEQ + HOME** (locked) both hands are free: black keys the sound, white keys its steps
SEQ + KNOB 1 (no step held)	the **sound** (drums; heard with DRUM PREVIEW ON, 2.7.1) / **note** to set
SEQ + KNOB 2	**DIV**: 1/4, 1/8, 1/16, 1/32, 8T, 16T, 1/2, 1BAR, 2BAR
SEQ + KNOB 3	**SWING** of the track
SEQ + KNOB 4	**LENGTH**: 1–64 steps
SEQ + step + KNOB 1	drums: which **sound** of the step LEVEL and RATCHET edit (heard with DRUM PREVIEW ON) · synths: the step's **note** (every note of a chord moves together)

Control	Does
SEQ + step + KNOB 2	drums: **LEVEL**: ghost, soft, norm, hard · synths (2.6): **VEL**, the step's velocity 1–127 (every note of the step; an accent is taken off, as it would play at 127)
SEQ + step + KNOB 3	**RATCHET**: x1–x4
SEQ + step + KNOB 4	**NUDGE**: –32 … +31 (in 1/64 of a step)
SEQ + step + PRESETS	a **parameter lock** on that step: the first click makes it one click away from the track's value, the next ones move it
SEQ + step + ALGORITHM	which parameter the lock is on
SEQ + step + OCT+	the step's **fill condition**: normal → FILL ONLY → NO FILL → normal
SEQ + step + OCT–	clears the step's nudge, locks and fill condition

A held step that you edited with a knob is **kept** when you let go (only a plain tap clears). Details in The step sequencer.

SEL — key and chords

Control	Does
SEL + any key	sets the **key of the song**: the root of all three synth tracks (*KEY D*)
SEL + KNOB 1	**CHORD** of the selected synth track: OFF, TRIAD, 7TH, 9TH, SUS4, POWER
SEL + KNOB 2	**SCALE** of all three synth tracks: 16 scales
SEL + KNOB 3	**KEYS**: OFF, SNAP, WHITE
SEL + KNOB 4	**TRANSPOSE** the selected track, ±24 semitones

Details in Key, scales, chords and CHORD+.

GLO — mix

Control	Does
GLO + key 1–4	**mute** / unmute track 1–4 (fades out in a few ms; the pattern runs on in time)
GLO + key 5–8	**solo** track 1–4 (several solos add up)
GLO + key 9 (held)	**fill** for as long as the key is held
GLO + key 10	**fill bar**: the whole next bar plays as a fill (press again before the bar to cancel)
GLO + key 16	**tap tempo** (two taps or more)
GLO + F#3 (the first black key, 2.6)	**MIDI LEARN** on / off; see MIDI LEARN
GLO + KNOB 1–4	the **levels** of tracks 1–4
GLO + SELECT	the tempo (also inside the visualiser)

The tiles show what is heard; the *fill* and *bar* tiles light while they act.

SAVE — song

Control	Does
SAVE + key 1–4	play section **A–D**: playing, from the next bar (every track from its first step); stopped, it becomes the loop at once

Control	Does
SAVE + key 1–4, several tapped while SAVE stays held	a **quick chain** (*A B B C*…, up to 8, repeats allowed): let go and they play in turn, round and round
SAVE + key 5–8	**save the loop** into section A–D (over a used section: press the key again within 3 s)
SAVE + key 13	switch **loop** / **song** mode
SAVE + key 14	**SONG REC**: from the next bar, every section you play (and how long) is written into the song; again (or STOP) to end
SAVE + key 16	the **SONG** screen

Details in Song mode.

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10. The step sequencer: steps, nudge, locks, fills

Each track has up to **64 steps** with its own **length** (1–64) and **division** (1/32 to two bars). Tracks of different lengths loop on their own and stay in phase (polymeters). A synth step holds up to 4 notes (a chord) with a level and ratchet, a tie or a rest, an accent and a slide; a drum step holds any of the 16 sounds, each with its own level and ratchet.

There are three ways to enter steps:

1. **Live** — REC and play (see Recording).
2. **The SEQ layer** — hold SEQ, the white keys are steps (all tracks; see SEQ — steps).
3. **The NOTES page** (synth tracks, 2.7.2; the STEP page before) — tap **SEQ**: the pattern as a **piano roll**, each note a bar as long as it lasts, the one you edit outlined in white with its name over it, the beats ruled, the C rows named on the left and the playhead a line. A note's **FORCE** shows as its fill: GHOST dark, SOFT hollow, NORM and HARD full, ACC a white cap; a SLIDE a dash to the next note.

NOTES page	Does
keys	the keys pressed together become the step under edit (POLY: up to 4 notes; MONO, LEG and UNI: the last one); in chord mode a key writes its whole chord, CHORD+ changes included; let go of all keys and it moves on to the next step (2.7.1: MENU → SEQ → **STEP ADVANCE OFF**: it stays, and the next keys write the same step again, for a chord tried twice) · **a MIDI keyboard** too (2.6): the notes it plays together on the selected track (the track its channel plays, GLO → MIDI) become the step, as they sound; on another track's channel, stopped, the screen says *MIDI PLAYS TRACK n* (and when you record: see REC)
KNOB 1 **NOTE**	goes from note to note: a chord's notes one by one, then the next step (the empty ones too, to put a note there); the TIEs of a note are its length, not stops
KNOB 2 **PITCH**	that note's pitch (a chord's others stay); an empty step gets your last note
KNOB 3 **LEN**	how many steps the note lasts: longer turns the steps after it into TIEs (it stops at the next step that has notes, and at the pattern's end: *NOTE AHEAD*, *END OF PATTERN*), shorter turns its last TIEs back into RESTs
KNOB 4 **FORCE**	**GHOST**, **SOFT**, **NORM**, **HARD**, then **ACC** (the accent: the step's, for its whole chord)
PRESETS	the zoom: 8, 16 (as it opens), 32 or 64 steps on the screen (the page with the note under edit)
EDIT (tap)	deletes the note under edit (a chord keeps its others; its last note takes the step and its TIEs)
OCT– / **OCT+** (tap)	the keys' octave, as everywhere
OCT+ held (0.4 s)	the note's **sheet** — (2.8.1) a strip along the bottom of the screen, so the piano roll stays visible; it reopens on the row you last used. KNOB 1 picks a row, KNOB 2 sets it, OCT+ does it, OCT– closes it — **SLIDE** (ON / OFF; OCT+ too), **NUDGE** (the step's micro timing, –32 … +31), **FILL** (NORMAL, FILL only, NO FILL), **MOVE** (KNOB 2: a step each detent, with its TIEs, nudge, fill and locks; not over another note: *NO ROOM*), **COPY** (OCT+: a copy just after it, picked, and MOVE next to place it), **DELETE** (OCT+)

EDIT + OCT– / OCT+ undo and redo the edits, as before. Not while recording or armed (the keys record live then). On the drum track, the NOTES page shows *DRUM TRACK*: use the DRUMS grid or the SEQ layer.

The DRUMS grid has the same sheet (2.7.2): on the grid, **OCT+ held** opens it for the hit of KNOB 1's sound on KNOB 2's step: **RATCHET** (x1 … x4), **NUDGE** and **FILL** (the step's: every sound in it), **MOVE** (KNOB 2), **COPY** (OCT+: to the next step), **DELETE** (OCT+); OCT– closes it. OCT+ tapped still turns the page of steps.

The **PATTERN** page (tap SEQ again) holds the track's **LEN**, **DIV**, **SWG** (swing) and **GATE** (how long each note lasts).

Nudge (micro timing)

SEQ + step + KNOB 4 moves a step off the grid, from –32 to +31 in 1/64 of a step: minus = early (the step plays before its grid time, inside the previous step), plus = late. Its ratchets move with it; recording and swing stay on the grid. On the drum track the whole step moves, every sound in it. A nudged step shows a dot in the corner of its tile.

Parameter locks

A **lock** gives one sound parameter another value **for that step only** (Elektron style).

1. Turn the parameter you want on its page (for example FX → DLY send, or LFO → RATE). That becomes the **lock parameter** (before you turn anything it is ENV DEST → FLT).
2. Hold **SEQ** and hold the step.
3. Turn **PRESETS**: the first click makes the lock, one click away from the track's value; the next ones move it. The title line shows it (*lock dst 14*; *lock flt* -- = no lock yet).
4. **ALGORITHM** (step still held) steps through the other parameters, wrapping round.

At the next step without a lock on that parameter, it comes back to its base value (notes still ringing follow). A knob turned on the page while a lock is in force wins: that value becomes the new base. Several locks can sit on one step (one per parameter), **24 per track**. The sound parameters of the track can be locked (and the PATTERN page's GATE); the arp, key / scale / chord, STRUM / VLEAD, voice-mode, MUTE, LEN / DIV / SWG and the global pages cannot (*NOT LOCKABLE*; *NO LOCK LEFT* when the 24 are used). A locked step shows the same dot as a nudged one; **SEQ + step + OCT–** clears both. Undo (EDIT + OCT–) is for steps and (2.7) their nudges: it does not bring locks back.

Fill conditions

SEQ + step + OCT+ cycles the step's condition:

Condition	Mark on the tile	Plays
normal	—	always
FILL ONLY	small **F** (top left)	only during a fill
NO FILL	**×** (top left)	always, except during a fill

Call a fill while you play: **GLO + key 9** (as long as you hold it) or **GLO + key 10** (the whole next bar). Build a beat whose rolls, crashes and pickup notes are FILL ONLY and whose main hat is NO FILL: one finger brings the fill in and out on the bar. A skipped step is silent whole (no note, no MIDI, no ratchet, no lock). On the drum track the condition is the step's, every sound in it. STOP ends any fill. The arp and the rolls ignore fills.

Divisions

DIV: 1/4, 1/8, **1/16** (default), 1/32, 8T, 16T (triplets), **1/2**, **1BAR**, **2BAR**. A 64-step track at 2BAR lasts 128 bars: drones, chord changes, slow songs.

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11. Key, scales, chords and CHORD+

Key and scale

- **SEL + any key** sets the **key of the song** for the three synth tracks. (Also SEL page → ROOT, per track.)
- **SCALE** (SEL + KNOB 2 sets all three synth tracks; the SEL page sets the selected track alone): CHR (chromatic), MAJ, MIN, DOR (dorian), MIX (mixolydian), PEN (major pentatonic), MPEN (minor pentatonic), HARM (harmonic minor), PHRY (phrygian), LYD (lydian), LOC (locrian), MEL (melodic minor), BLUES, WHOLE (whole tone), DIMHW and DIMWH (diminished, half-whole and whole-half).
- **KEYS**: **OFF** (chromatic), **SNAP** (every key rounded down to the scale), **WHITE** (the white keys walk the scale from C4, the black keys are silent).
- **TRANSPOSE** (per track): ±24 semitones.

Changing a sound (PRESETS, a user preset) never changes the key, the chord mode, the pattern or the mix of its track.

Chord mode: one finger, one chord

Hold **SEL** and turn **KNOB 1 CHORD** on the selected synth track:

CHORD	Notes
OFF	single notes
TRIAD	1-3-5
7TH	1-3-5-7
9TH	1-3-7-9 (the lo-fi / R&B voicing)
SUS4	1-4-5
POWER	1-5-8 (root, fifth, octave)

With a chord on, **the white keys walk the scale from C4**: C4 plays the chord of the key's I, D4 the II, E4 the III… and the keys below C4 carry on downwards. Every chord is built from the scale, so it always fits the song. With SCALE on CHR, the chords come from the minor scale. One finger plays the whole chord, and it is recorded as a chord.

CHORD+: the black keys change the chord

With a chord mode on, **hold a black key while you play a white one** — or press it while the chord is held, and the chord changes under your finger:

Black key	Changes the chord
F#	major ↔ minor
G#	adds the 7th
A#	sus4
C#	adds the 9th
D#	an inversion

Both octaves of black keys work, and you can hold several. The 7th and 9th come from the scale: in C major, G4 plays **G** (the V chord), **G#** makes it **G7**, **F# + G#** make it **Gm7**; on C4, F# + G# give C-E♭-G-B. What you play is recorded as it sounds.

SEL 2: STRUM and VLEAD

Tap **SEL** twice (or SEL, then SELECT) for **SEL 2**:

Knob	Does
KNOB 1 **TRN**	transpose the track, ±24 semitones
KNOB 2 **STRUM**	plays a chord's notes one after the other like a strummed guitar, 1–60 ms a note: right = low to high, left = high to low, centre = off. On the keys and on the chord steps the sequencer plays
KNOB 3 **VLEAD**	**ON**: voice leading — each chord is voiced nearest the last one, so a progression moves smoothly instead of jumping

12. Mix: levels, mute, solo, fills, tap tempo

Where	What
GLO + KNOB 1–4	levels of tracks 1–4
TRACKS screen, KNOB 2	the selected track's level (on a muted track: unmutes it)
GLO + keys 1–4 / 5–8	mute / solo
VOICE 2 page	PAN and MUTE of a synth track
TRACKS screen, KNOB 4	the selected track's pan
GLO → DRUMS	the drum track's LVL, its reverb and (2.5) delay sends, and its MIDI channel
MASTER knob	the output volume

A muted track fades out in a few milliseconds and plays no new notes; its pattern runs on in time. Several solos add up.

Tempo: SELECT on TRACKS (or inside any layer), **GLO + key 16** tap tempo, or GLO → GLOBAL → BPM (40–240). With MIDI SYNC on, the tempo follows the master.

13. Undo, clear, save, sections, autosave

Action	How
Undo / redo	**EDIT + OCT–** / **EDIT + OCT+**. One level: the last recording pass, erase, clear, step or pattern edit
Clear the selected track	**hold REC** ~2 s (after 0.7 s a ring fills; let go before it is full: nothing). Undo brings it back
Clear the pattern	SAVE → TOOLS → **CLRSQ** (GO)
Reset the sound	SAVE → TOOLS → **INIT** (GO): the engine's defaults and first preset
New loop	SAVE → TOOLS → **NEW** (GO): the four tracks of your working loop back to their power-on sounds, empty patterns, 90 BPM (*NEW LOOP, A-D KEPT*: the saved sections are not touched; stop the song first in song mode; undo brings back only the last track's pattern, so save a section first if in doubt)
Save the loop as a section	**SAVE + key 5–8** (A–D)
Save / load a section	SAVE → SECTION: **SECT** A–D, **LOAD** (GO), **SAVE** (GO)
Erase a saved section (2.6)	SAVE → SECTION: **SECT** A–D, **ERASE** on KNOB 2 (GO), the song stopped: the slot is *EMPTY* again and its section key goes dark. The steps of the song that played it are taken out (the others keep their order; none left: the default song A B C D). Your working loop is not touched. No undo: save it into another slot first if in doubt
Save a sound	SAVE → USER: **SLOT** (1–32), **SAVE** (GO)
Autosave	automatic: when the transport is stopped and you have not touched anything for 2.5 s (at most every 20 s), the working project is kept in flash; at power-on SLOOP comes back exactly as you left it

One song on the FM-1. The FM-1 holds one song: **four sections, A–D**, and the order they play in (the SONG screen). A **section** is a loop: the four tracks (sounds, patterns, kit, locks, nudges, fill conditions, the track filter) and the global settings of the loop (tempo, swing, key…). Your **working loop** (what plays in loop mode, kept by the autosave) is apart from them: LOAD copies a section into it, SAVE copies it into a section. A–D are not four songs. **To keep several songs**, save an editor **Backup** file per song (Song & sections tab) and restore the one you want to play. (Before 2.7 the SAVE page was called PROJECT, its slots 1–4: the same four sections.)

Settings of the FM-1 — SYNC, MIDI OUT, IN, the lights, the menu settings, the REC screen's MODE, START and QUANT — are not part of a section: loading one or NEW never changes them.

The SAVE pages (SAVE tapped, not on TRACKS)

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
PRESETS	the sound (your favourites, then factory by kind, then user presets)	the engine (next / previous)	—	**FAV** (2.7): right adds the sound to your favourites, left removes it
USER	**SLOT** 1–32	**LOAD** (GO)	**ERASE** (GO)	**SAVE** (GO)
SECTION	**SECT** A–D	**ERASE** (GO, 2.6)	**LOAD** (GO)	**SAVE** (GO)
TOOLS	**CLRSQ** (GO)	**INIT** (GO)	—	**NEW** (GO)

Favourites (2.7). On the PRESETS page, turn KNOB 4 right: the sound is added to your favourites (*FAV ADDED*), up to 16, which come first in the PRESETS list, tagged FAV, in the order you added them; the PRESETS knob reaches them in a click or two from anywhere on TRACKS or HOME. KNOB 4 left removes it. A user preset can be a favourite too. They are a setting of the FM-1, kept through power-off (not in a project, and not in an editor backup). MIDI Program Change still counts the list without them.

GO: one click shows *AGAIN: SAVE* (or LOAD…), a second click within about 1.5 s does it. Presets and sections are written to flash only while the song is stopped.

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14. Song mode: sections, chains, songs

The FM-1 holds **one song**: up to **16 steps** of **4 sections, A–D**. Each section holds the four tracks (sounds, patterns, kit): SAVE → SECTION shows them. For more songs, keep one editor backup file per song.

Make it live, by playing

1. Make a loop (the verse). **SAVE + key 5**: *save A*. Change the loop (the chorus) and save it into **B** (key 6), a bridge into **C** (key 7), an ending into **D** (key 8). Saving over a used section asks for the key again within 3 s.
2. **Play the sections live: SAVE + key 1–4.** Playing: the section starts on the next bar, every track from its first step, always in time. Stopped: it becomes the loop at once.
3. **Quick chain:** keep SAVE held and tap several section keys — *A B B C*, any order, repeats allowed, up to 8 — then let go. The first plays on the next bar, then each next one after the previous has played its **pattern length** (the longest track's), round and round. The title line reads *chain A B B C*; the section playing is lit, the next one framed. A single section tap, STOP, or PLAY in song mode ends the chain.
4. **Record the song as you play it: SAVE + key 14.** From the next bar every section you play, and for how many bars, is written into the song. SAVE + key 14 again, or STOP, ends it (*SONG PARTS 5*). It is saved by itself once you stop. SONG REC records what a chain plays too.
5. **Play it back: SAVE + key 13** switches *loop* / *song*. In song mode **PLAY** plays the whole song and stops at the end (your loop is back afterwards).

The SONG screen

Open it with **SAVE tapped on TRACKS** (or on DRUMS), or **SAVE + key 16**. It shows the chain and edits it by hand:

Control	Does
KNOB 1	the song step (cursor)
KNOB 2	that step's section (A–D)
KNOB 3	its length in bars (1–64)
KNOB 4	the number of steps in the song (1–16)
REC	stores the current loop into the step's section (a used section: REC again within 3 s)
SAVE (tap)	saves the song
OCT–	loop / song mode
OCT+ twice (within 3 s)	loads the step's section (unsaved work is replaced)
PLAY	play / stop
SEQ	the SEQ pages
HOME	back to TRACKS

The song's settings can only be edited while it is stopped (*STOP FIRST*). An empty section cannot play (*EMPTY SECTION: REC*).

15. The screens

Screen	How to get there	What it shows · what the knobs do
TRACKS	tap **HOME**	the performance view: tempo, swing, transport, bar.beat; each track with its sound, steps, playhead, mute / solo / rec badges and level. **(2.8)** a coloured **A–D** badge by the title shows which section is playing. Dials: KNOB 1 **swing** (global) · KNOB 2 **level** (selected track; on a muted track: unmutes it) · KNOB 3 **steps** (its length) · KNOB 4 **pan**
Visualiser	tap **HOME** on TRACKS	twelve full-screen visualisers; see The visualiser
Layer	hold a layer button	16 tiles (the white keys) and the four dials, in the layer's colour
DRUMS (grid / kit)	tap **EDIT** or **SEQ** on TRACKS with the drum track	see The DRUMS screen
REC READY / FREE TAKE / COUNT-IN	**REC** while stopped	the tracks, then **mode · length · start · quant** on KNOB 1–4 (4-3-2-1 during a count-in); in a free take: the seconds and the loop it would make
Hold ring	hold **REC**	the ring of *clear track* filling
SONG	tap **SAVE** on TRACKS, or **SAVE + key 16**	the section chain; see The SONG screen
Pages	tap ENV, LFO, FX, SEL, EDIT, ARP, SEQ, GLO, SAVE	four colour-coded values with a graph (ENV, LFO, ARP, SEL, FX, SLICER…), or the four values in large type placed as the knobs are (1 2 / 3 4)
Menu	hold **HOME**	the settings of the FM-1 in four sections; see The HOME menu

The **title line** of every screen also carries short messages: *KEY D*, *ERASED*, *AGAIN: SAVE*, *NEXT: B*, *chain A B B C*, *lock flt 12*…

The lettering (2.6) is Terminus **Bold**, all in **capitals**: thicker strokes, easier to read on the small screen at arm's length; the layout is the same as before (same 8 × 16 cell).

16. Every page, knob by knob

Tap a button to open its first page (or the last one you used); **tap it again** for its next page, or turn **SELECT** to go back and forth. The number after the title (*PATTERN 2/2*) counts the pages SELECT goes through; tapping SEQ once more also opens the SONG screen, a screen of its own (2.7: it is no longer counted, so SEQ no longer reads *2/3*). On the drum track, pages that do not apply to it show *DRUM TRACK*.

Button	Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
ENV	**ENV**	ATK attack	DEC decay	SUS sustain	REL release
	ENV DEST	FLT: envelope → brightness (±)	PIT: envelope → pitch (±, the 808 punch)	SHP: envelope → the engine's shape (±)	ESYNC (2.6): OFF / ON, ATK DEC REL as note values
LFO	**LFO**	RATE (Hz)	WAVE: SIN, TRI, SAW, SQR, S&H	PHS phase	FADE (fade-in time)
	LFO 2 (2.6)	SYNC: OFF, 4BAR, 2BAR, 1BAR, 1/2, 1/4, 1/8, 8T, 1/16, 16T, 1/32	TRIG: NOTE / FREE	POL: BI / UNI	—
	LFO DEST	PIT: vibrato	FLT: filter wobble	SHP: shape	AMP: tremolo
FX	**FX** (sends)	DST drive	CHO chorus send	DLY delay send	REV reverb send
	FILTER	FILT: one-knob filter of the track — left LP, right HP, centre OFF (the drum track too; 2.7.2: its page no longer greyed)	—	—	—
	SLICER	SLCR: OFF, GATE, STUT (stutter)	PAT: pattern 1–16	RATE: 1/8, 1/16, 1/32, 8T, 16T, 32T	DEPTH
	INSERT (2.6)	INSRT: OFF, SOFT, HARD, FOLD, FUZZ, CRUSH, PHASR, FLANG, CHOR	A	B	C (what A B C are depends on the type: see Per track)
	INSERT 2 (2.6)	MIX: dry / wet	—	—	—
	DLY (global)	TIME: 1/4, 1/8, 1/16, 1/32, 8T, 16T, 1/8D, 1/16D, 1/4D (2.6; at 60 BPM and below it stops at 1.49 s)	FDBK feedback	COLR colour (tone of the repeats)	MIX
	REV/CHO (global)	SIZE reverb size	DAMP reverb damping	CRT chorus rate	CDP chorus depth
SEL	**SEL**	ROOT (the track's key)	SCL: the track's scale (SEL + KNOB 2 sets all three)	QNT keys: OFF, SNAP, WHITE	CHORD: OFF, TRIAD, 7TH, 9TH, SUS4, POWER
	SEL 2	TRN transpose ±24	STRUM ±60 ms	VLEAD: OFF / ON	—
EDIT	**EDIT 1**	engine parameter 1	2	3	4 (see the engines)
	EDIT 2	engine parameter 5	6	7	8
	FILTER (ANALOG only, 2.6)	TYPE: LP, BP, HP	CUT cutoff	RES resonance	KTR key tracking

Button	Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
	VOICE	VCE: POLY, MONO, LEG (legato), UNI (unison)	GLD glide	GLMOD: RATE / TIME	PRIO: LAST, LOW, HIGH (mono note priority)
	VOICE 2	ALLOC: ROT (rotate) / REUSE	DTUNE (unison detune)	PAN	MUTE
ARP	**ARP**	MODE: OFF, UP, DN, UPDN, RND, ORD, and (2.7) REPEAT, DNUP, UP+8, CONV, DIVG, PINKY, THUMB, WALK, CHORD (below)	RATE: 1/4 … 16T	OCT: 1–4 octaves	GATE
	ARP 2	SWG swing (2.7: also with the sequencer stopped)	PROB probability of each note	HOLD: OFF / ON (latch; with MODE OFF, 2.7: KBD HOLD)	ORD: NOTE (pitch order) / PLAY (the order you played)
SEQ	**STEP**	STEP (cursor)	NOTE	TIME: NOTE, TIE, REST · on a note: LEN (2.6)	FLAG: –, ACC, SLD, A+S
	PATTERN	LEN 1–64 steps	DIV 1/4 … 2BAR	SWG track swing	GATE note length
	SONG	the SONG screen			
GLO	**GLOBAL**	BPM 40–240	SWING (all tracks)	CLICK: OFF, REC, ON	TUNE ±50 cents
	MASTER	DUST	DUCK	FILT (DJ filter)	ROLL (note-repeat rate)
	SYSTEM	OUT (MIDI out): KEYS / SEQ (also on MIDI 2)	SYNC: INT, USB, TRS	IN: NOTES / CLOCK / CH 1–16 (also on MIDI 2)	CPU load (USB link state when no computer is connected)
	DRUMS	CH4: the drum track's MIDI channel (OFF, 1–16, default 10; the same setting as GLO → MIDI → CH4; 2.6: a setting of the FM-1)	LVL drum level	REV drum reverb	DLY drum delay (2.5): the drums into the tempo delay (FX → DLY sets its time, feedback and tone), saved with the project
	TEXTURE (2.6)	SRC: OFF, VINYL, TAPE, RAIN, WIND, RADIO, ROOM, USR1–4	LEVEL	TONE (LP ← OFF → HP)	MOVE
	TEXTURE 2 (2.6)	RATE: 1–16 bars	WHEN: PLAY / ALWAYS	DUCK: OFF / ON	—
	MIDI (2.6)	CH1: track 1's MIDI channel (OFF, 1–16)	CH2: track 2's	CH3: track 3's	CH4: the drum track's (default 10)
	MIDI 2 (2.6)	IN: NOTES / CLOCK / CH 1–16	OTHER: the channels no track has: SEL (the selected track) / OFF	OUT: KEYS / SEQ	BEND: the pitch bend's range, 1–12 semitones
	MIDI 3 (2.7)	FX CH: the punch-in effects' MIDI channel (OFF, 1–16)	DR MAP: how MIDI notes play the drums (GM / 1-16)	REMOTE: the buttons over MIDI (OFF, 1–16)	CLOCK: OUT, the MIDI clock and START / STOP out on USB (OFF / OUT)
SAVE	**PRESETS**	the sound	the engine	—	—
	USER	SLOT 1–32	LOAD (GO)	ERASE (GO)	SAVE (GO)

Button	Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
	PROJECT	SLOT 1–4	ERASE (GO)	LOAD (GO)	SAVE (GO)
	TOOLS	CLRSQ: clear the pattern (GO)	INIT: reset the sound (GO)	—	NEW project (GO)

Notes:

- **Global** pages (DLY, REV/CHO, GLOBAL, MASTER, SYSTEM, DRUMS) are the same whatever track is selected.
- **SYNC**, **MIDI** and **IN** are settings of the FM-1, kept when you load a project.
- On the **drum track**, the pages that work are SLICER, PATTERN (SEQ), the SONG screen and the global pages; its kit, level, reverb and pan are on the DRUMS kit page, and its filter is **FX + KNOB 4**.
- FM6 tracks ignore the ENV page and ENV DEST (each operator has its own envelope).
- **ARP modes** (2.7, after Felucca 1.4 / 1.5.1). With C, E and G held: **UP** C E G · **DN** G E C · **UPDN** C E G E · **DNUP** G E C E · **RND** at random · **ORD** in the order you pressed them · **REPEAT** the last note you pressed, again and again (OCT ignored) · **UP+8** C E G C' (the first note again an octave above the top) · **CONV** C G E (from the outside in) · **DIVG** E G C (from the inside out) · **PINKY** C G E G (the top note between the others) · **THUMB** C E C G (the bottom note between the others) · **WALK** one step up or down at random, from the bottom · **CHORD** the held notes together (the first four), on every step, an octave higher each step through OCT. OCT spreads the list over 1–4 octaves; ARP 2 ORD PLAY keeps the order you played for the other modes. Until 2.6, ORD played as UP unless ARP 2 ORD was PLAY: now it always follows the order you pressed.
- **KBD HOLD** (2.7): ARP 2 → **HOLD ON** with ARP → **MODE OFF** holds the notes you play, without an arpeggio: let go and they go on sounding (a pad, a drone, a chord to play over with one hand). The next note you play with no key down **replaces** them (a new chord); notes played while keys are still down **join** them. The keys' MIDI note-offs wait too, so gear played from the FM-1 holds with it. **HOLD OFF** (or ARP on, another project, NEW) ends them; loading another sound ends the held notes too, but HOLD stays ON. Projects saved before 2.7 open with HOLD OFF when their arp is OFF (it did nothing then), so nothing hangs by surprise. It works for MIDI notes too. Recorded as you hear it: a held note ends where it is replaced. Up to 2.6, HOLD did nothing with the arp OFF.
- **ARP SWG** (2.7): the arp's swing also works with the sequencer stopped (the odd steps late), as it does while playing; until 2.6 it only worked while playing.
- **ARP and sounds** (2.7): while the arpeggiator runs (MODE not OFF), loading another sound (a preset or a user preset) keeps the arp: its settings (both ARP pages) and the notes it holds or latches go on with the new sound. With the arp OFF, a preset that has an arp of its own (8BIT ARP…) brings it, as before. (2.7.1) An arp that came with the sound (8BIT ARP, a user preset saved with its arp) and that you did not change goes with it: the next sound brings its own (most often OFF). Change one of its settings and it is yours: it goes on with the next sounds.
- **ENV SYNC** (ENV DEST → ESYNC ON, 2.6): ATK, DEC and REL become note values of the tempo, from 0 through 1/64T … 1/4, 1/2 … to 4BAR, with triplets and dotted values; at 120 BPM an ATK of 1/4 lasts half a second, at 60 a whole one. They follow the tempo as it changes. OFF: times in milliseconds, as before.
- **LFO 2** (2.6): **SYNC** locks the LFO's cycle to a note value of the tempo (RATE is then ignored; OFF: RATE, as before) · **TRIG** NOTE restarts the LFO at PHS on each new phrase (as before), FREE lets it run on · **POL** BI swings above and below the value, UNI only above it (a tremolo that only lowers the level, a wobble that only opens).
- **FILTER** (EDIT, ANALOG only, 2.6): the ANALOG engine's filter as a **low-pass** (as before), **band-pass** or **high-pass**; CUT, RES and KTR are the same values as on EDIT 2.
- On the screen, EDIT 1 and EDIT 2 carry the engine's own page names: ANALOG *OSC* / *FLT*, DIGITAL *OPS* / *MOD*, PHASE *PHS* / *LINE*, LOFI *CHIP* / *MOTN*, SAMPLE *SET* / *TONE*, VOICE *VOWL* / *TONE*, TRIO *OSC* / *TONE*, WHEEL *BARS* / *TONE*, GRAIN *GRAN* / *SPRY*, FM6 *OPS* / *PATCH*, PHYS *BODY* / *EXCT*, NOISE *SRC* / *MOVE*.

- Any sound parameter on these pages can be **locked** on a step (see Parameter locks), except the pattern, arp, key and voice-mode ones.

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17. The sounds and the twelve engines

The sound bank

153 factory sounds, every one a full patch on one of the twelve engines, all level-matched (the same LEVEL gives the same loudness). **PRESETS** browses them **by kind** — basses, keys, organs, pads, leads, plucks and bells, stabs, FX — with the kind shown next to the name; your **32 user presets** come after. A new project starts at **90 BPM** with *808 BOOM* on track 1, *RHODES* on track 2, *LOFI FLUTE* on track 3 and the 808 kit on track 4.

Kind	Sounds (engine)
Bass	808 BOOM, 808 DIRTY, 808 SLIDE, SUB BASS, PLUGG BASS (they slide between held notes, two octaves under the keys) · REESE, WOBBLE, ACID 303, FUNK BASS, MONO BASS (ANALOG) · FM BASS, BASS GTR (DIGITAL) · CZ BASS (PHASE) · ROUND BASS (FM6) · FAT BASS (TRIO) · WOW BASS (VOICE) · GB BASS (LOFI) · UP BASS, DEEP BASS (SAMPLE) · PLUCK BASS (PHYS) · NES BASS (LOFI) · SAW BASS (ANALOG)
Keys	RHODES, DX RHODES, WURLI, M1 PIANO, AFRO KEYS, CLAV (DIGITAL) · GRAND PNO, DUSTY PNO, LOFI KEYS (SAMPLE) · SOFT KEYS (PHASE) · TINE EP (FM6) · FM GRAND (DIGITAL)
Organ	SOUL ORGAN, GOSPEL, JAZZ ORGAN, DIRTY B3, HOUSE ORGN (WHEEL) · DRAWBARS (FM6) · ACCORDION (TRIO) · ROCK ORGAN, CHAPEL, REGGAE ORG (WHEEL)
Pad	WARM PAD, DARK STR, ATMOS PAD (ANALOG) · SAW PAD, ENSEMBLE (TRIO) · GLASS PAD (DIGITAL) · SOFT PAD (FM6) · CZ STRING (PHASE) · LOFI CLOUD, VIBE HAZE, STR CLOUD, SHIMMER, DRONE (GRAIN) · CHOIR AAH, SOUL OOH (VOICE) · BOWED MTL, CELLO BOW, GLASS BOWL (PHYS) · WIND, OCEAN (NOISE) · CZ PAD (PHASE) · VOX PAD (VOICE) · PNO FREEZE, HORN CLOUD (GRAIN) · SINE PAD (ANALOG) · PULSE PAD (TRIO)
Lead	SUPERSAW, G-FUNK LD, FAT LEAD, PAN FLUTE (ANALOG) · SYNC LEAD, HOOVER (TRIO) · TALKBOX, HARMONICA (VOICE) · GAME LEAD (LOFI) · LOFI FLUTE (SAMPLE) · FLUTE DUST (GRAIN) · NZ ARCADE (NOISE) · WAVE LEAD (LOFI) · CZ LEAD (PHASE) · ROBOT (VOICE) · PWM LEAD (ANALOG) · SQR LEAD (TRIO)
Pluck & bell	TRAP PLUCK (ANALOG) · RESO PLUCK (PHASE) · PLUGG BELL, TRAP BELL, MUSIC BOX, KALIMBA, MARIMBA, XYLOPHONE, GLOCKEN, CELESTA, STEEL DRUM, HARP (DIGITAL) · VIBES (SAMPLE) · 8BIT ARP (LOFI) · GLASS BELL, WOOD BARS, NYLON PICK (FM6) · STR PLUCK, SITAR, PHYS HARP, BELL TREE, MODAL BAR, THUMB PNO, NYLON GTR, STEEL GTR, MUTED GTR, KOTO, BANJO, TANPURA, CHIMES, BIG BELL, VIBRA BAR, STEEL PAN, WOOD BLOCK (PHYS) · 1BIT BEEP (LOFI) · CZ RESO (PHASE) · TUBE BELL, FM PLUCK (DIGITAL)
Stab	MIN STAB, MIN7 STAB, RAVE STAB, DUB CHORD (TRIO: one key plays the chord) · SYN BRASS (ANALOG) · CZ BRASS (PHASE) · BRASS SECT (FM6) · 80S POLY (TRIO) · HORN STAB, STRING STB (SAMPLE) · CHIP CHORD (LOFI) · POWER STAB (TRIO)
FX	SCRATCH (scratch, backspin, rewind across the keys) · GM KIT (SAMPLE) · HAND DRUM, MEMB TOMS, TABLA, CONGA, TIMPANI (PHYS) · RAIN, NZ METAL, VINYL, HISS, RISER, NZ SNARE, BITCRUSH, RADIO (NOISE) · CHIP NOISE (LOFI)

To pick an engine directly: SAVE → PRESETS → **KNOB 2**. To start from the engine's defaults: SAVE → TOOLS → **INIT**.

The twelve engines

Each engine's eight parameters are on **EDIT 1** (KNOB 1–4) and **EDIT 2** (KNOB 1–4), tap EDIT once and twice; the screen shows them under the engine's own page names (for example DIGITAL: *OPS*, *MOD*). A dash is an unused knob.

Engine	Sound	EDIT 1: KNOB 1 · 2 · 3 · 4	EDIT 2: KNOB 1 · 2 · 3 · 4
ANALOG	virtual analogue: two oscillators, filter	WAVE (SAW, SQR, TRI, SIN, PWM) · DTN detune · MIX · NOIS noise	CUT cutoff · RES resonance · DRV drive · KTR key tracking

Engine	Sound	EDIT 1: KNOB 1 · 2 · 3 · 4	EDIT 2: KNOB 1 · 2 · 3 · 4
DIGITAL	4-operator FM	ALG algorithm 1–8 · R2 · R3 · R4 (operator ratios)	IDX FM amount · MDEC modulator decay · FB feedback · —
PHASE	phase distortion (CZ style)	WAVE · WAVE2 (a second wave, or –) · DCW (the distortion amount) · ENV (its envelope)	DTN detune · LINE: MIX / RING · SUB sub-oscillator · —
LOFI	8-bit chip	CHIP: 4BIT, 4B/2, 8BIT, 1BIT, STEP · WAVE: PLS, TRI, SAW, NOIS, WRAM · DUTY (pulse width / wave table) · CRSH crush (DCY on STEP)	SWP pitch sweep · VIB vibrato · ARP: OFF, OCT, MAJ, MIN (chip arpeggio) · TONE
SAMPLE	sampler	SET: PIANO, BASS, VIBES, HORNS, STRGS, FLUTE, SCRCH (2.7.2: one scratch over the whole keyboard), PERC, **USR1–USR4** · TUNE ±24 · BITS (bit reduction) · LOOP: OFF / ON	CUT (tone) · — · DRV drive · —
VOICE	formant voice (choirs, talkbox)	VOWL vowel · VOWL2 second vowel · TALK (time from one to the other) · SHIFT formant shift ±12	BUZZ · BRTH breath · Q · RAND
TRIO	three oscillators	WAVE (16 combinations) · INT2 · INT3 (intervals of oscillators 2 and 3, ±24) · DTN	MODE: LP, BP, HP, NOT (filter type) · CUT · RES · PW pulse width
WHEEL	tonewheel organ	REG (16 drawbar registrations: FLUTE, MELLO, HOLLW, SMOOT, 3BAR, BLUES, GOSPL, ROCK…) · SUB · BODY · TOP (drawbar groups ±8)	PERC: OFF, 2ND, 3RD, 2SOFT, 3SOFT, 2SLOW, 3SLOW · CLICK · DRV · ROTR: OFF, SLOW, FAST (rotary speaker)
GRAIN	granular	SRC (any sample set, USR1–USR4 too) · POS position · SIZE grain size · DENS density	PTCH ±24 · SPRD spread · RAND randomness · TONE
FM6	six-operator FM, DX7 patches	ALG: PAT (the patch's) or 1–32 · FB feedback · MLVL modulator level (brightness) · MRAT modulator ratio	MEG modulator envelopes slower / faster · VMOD velocity → brightness · DTUN carrier spread · PTCH: F1–F8 factory, B1–B27 your bank
PHYS	physical models: strings, bars and bells, drum heads, sympathetic strings (DaisySP and Rings, as Felucca 1.0 ported them; 3 voices a track, SYMP 2)	MODEL: MODAL, STRNG, MEMB, SYMP · STRC structure (MEMB: HARM, SYMP: CHRD the sympathetic strings' chord) · BRIT brightness · DAMP	POS strike / pluck position (SYMP: SYMP, how much they ring) · ACC accent · BOW bowing (MEMB: BEND, SYMP: BUZZ) · EXC the exciter in the mix
NOISE	noise, from analogue to digital: colours, crackle, shift registers, metallic tones	MODE: ANLG, DUST, LFSR, META · COLR white → pink → brown · FREQ · RES	TRK key tracking · DENS impulses (LFSR, META: LEN, the register) · DRFT drift · CRSH crush (LFSR, META: CLK, the clock above the key)

The **SAMPLE** sets are free recordings (CC0: Versilian Studios VSCO-2 CE and VCSL, Sonic Pi), coloured like a record through an old sampler. **GRAND PNO** is a Steinway recorded note by note; long notes fade as on the real one.

Polyphony

The three synth tracks **share one pool of 8 voices**; the drum track has its own 6. Played on its own, each engine's maximum polyphony is:

Voices	Engines
8	ANALOG, DIGITAL, PHASE, LOFI, SAMPLE, TRIO, WHEEL, NOISE
6	FM6
4	VOICE
3	GRAIN, PHYS (SYMP model: **2**)

In **MONO** or **LEGATO** a track plays one note at a time; **UNISON** stacks detuned voices within the same cap. When more than one synth track sounds at once, the 8-voice pool is split between them: a part that needs a voice takes the oldest sounding one.

FM6 and DX7 patches

FM6 is msfa, the synthesis core of Dexed, as Felucca 1.0 ported it: six operators with their own four-stage envelopes, keyboard scaling, velocity, ratio or fixed frequency and detune; 32 algorithms, feedback, LFO, pitch envelope; six voices per track. The eight EDIT values above are **macros on top of the patch**; the track's FLT moves MLVL (so LFO → FLT still brightens it), SHP the feedback, PIT the pitch.

Edit full patches in the web editor's **FM6** panel (Sound page, on an FM6 track): every operator, **Send to track**, **Store in bank** (B1–B27), **Import SysEx** (a DX7 voice or a 32-voice bank: thousands of DX7 and Dexed patches play on the FM-1; or drop the .syx file on the panel) and **Export SysEx**. **Store cartridge in bank** (2.5) puts a whole 32-voice cartridge on the FM-1 at once: the first 27 voices are ticked (or tick the ones you want, up to 27), and one write puts them in B1, B2… in order; the editor asks before replacing, and the rest of the bank stays. Stop the song first. Before storing you can choose to **also add them to the user presets** (in the free slots, or from a slot you pick): each voice gets a user preset with its name that plays its B slot, so on the FM-1 you pick it by name with **PRESETS** or **SELECT**, like any sound; the editor says which user presets it would replace. (A user preset keeps PTCH, not the patch: the voice itself stays in the bank.) Then pick them on the FM-1 with **PTCH** (EDIT page 2: B1–B27 after the factory F1–F8). A project keeps the track's **PTCH**, not the patch itself: store an edited patch in the bank to keep it.

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18. Effects: master, punch-in, per track

Signal path

Each synth track: engine → **drive** (DST) → **INSERT** (2.6) → **SLICER** → **FILTER** (the track's own) → level (and DUCK) → pan and **sends** to the stereo chorus, the tempo delay and the reverb. The drum track has its own INSERT (2.6), slicer, filter, level, pan, reverb send and (2.5) delay send. Then the whole mix, with the effect returns (and, 2.6, the **TEXTURE** bed): **DUST** → **punch-in effects** → **FILT** (the DJ filter) → **MASTER** volume → limiter (2.6: it looks one block, 0.7 ms, ahead: a loud hit is already turned down when it comes, never pushed into the output's soft clipper).

Master: DUST, DUCK, FILT, ROLL

On FX + KNOB 1–3, or GLO → MASTER:

	Does
DUST 0–100 %	the mix through an old sampler and a record: drive into a soft clip, a lower sample rate (down to ~11 kHz), fewer bits (down to 8), a low-pass closing to ~3 kHz, and, while the transport plays, a little hiss and crackle (a stopped SLOOP is silent)
DUCK 0–100 %	every kick pumps the synth tracks down and back over an 1/8 note: the sidechain sound, in time at any tempo
FILT	a DJ filter: left of centre a low-pass closing, right a high-pass opening, centre OFF; it glides (no zipper noise)
ROLL	the note-repeat rate of ARP + key

Punch-in effects

FX + white key: the effect runs on the whole mix while the key is held and lets go cleanly when you release it. Loops and the gate are locked to the tempo and start on the grid. **Fade (2.8):** hold **FX**, then **SELECT** (shown top right) sets a fade-in and out for the punch-ins, 0 (off — instant, just an anti-click) to 15 (about 0.37 s): the effects ease in and out instead of snapping on and off. **Over MIDI (2.7):** set GLO → MIDI 3 → **FX CH**, and that channel's notes F3 … G5 play the same 27 effects (see the MIDI settings).

Key	Effect	Key	Effect
1 · F3	loop 1/4	9 · G4	low-pass sweep
2 · G3	loop 1/8	10 · A4	high-pass sweep
3 · A3	loop 1/16	11 · B4	phone
4 · B3	loop 1/32	12 · C5	bit crush
5 · C4	stutter (1/16 triplets)	13 · D5	alias (sample-rate drop)
6 · D4	reverse	14 · E5	gate 1/16
7 · E4	tape stop	15 · F5	echo (dotted 1/8)
8 · F4	half speed	16 · G5	tape wobble

FX + black key (2.6): eleven more, on the black keys from the lowest F# to the highest.

Key	Effect	Key	Effect
F#3	**KILL LOW**: the lows gone (below ~250 Hz, 24 dB/oct, DJ style)	F#4	**FLANGER**: a jet sweep over a bar, left and right apart
G#3	**KILL MID**: the mids gone (~250 Hz – 2.5 kHz)	G#4	**AUTO-PAN**: the mix in mono, left and right once a beat
A#3	**KILL HIGH**: the highs gone (above ~2.5 kHz)	A#4	**FREEZE**: the mix poured into the reverb and held there for as long as you hold the key; let go and it dies away with the reverb's SIZE
C#4	**REWIND**: the record pulled back: it slows, stops and spins backwards faster and faster (about a third of a second of the past), then silence until you let go	C#5	**LOOP 8T**: loop 1/8 triplet
D#4	**OCT UP**: the mix an octave up	D#5	**MONO**: left and right in one
		F#5	**FADE OUT**: the mix fades to nothing over two beats

The screen shows the 27 keys as a keyboard: the white keys in four rows of four, as on the other layers, the black keys above them, each between the two white keys it sits between on the FM-1 (the one after a row's last white key at the right end of that row). The effect held is lit white.

Texture (2.6)

GLO → TEXTURE (and TEXTURE 2): a looping bed under the whole mix, the glue of lo-fi and ambient tracks, without taking a track. It goes into the dry mix (not the sends), before DUST, the punch-ins and the master filter.

Knob	TEXTURE	TEXTURE 2
1	**SRC**: OFF, **VINYL** (hiss, rumble, crackle), **TAPE** (dark hiss, drifting), **RAIN** (a bed and droplets), **WIND** (resonant gusts), **RADIO** (bursts of static), **ROOM** (room tone), **USR1–USR4** (your sample slot, its first sound looped at its own pitch)	**RATE**: how long MOVE's breath takes: 1, 2, 4, 8 or 16 bars of the tempo
2	**LEVEL**	**WHEN**: PLAY (it fades in and out with the transport) / ALWAYS
3	**TONE**: left a low-pass, right a high-pass, centre open	**DUCK**: ON, the kick pumps it too (GLO → MASTER → DUCK)
4	**MOVE**: how much it breathes over RATE (its level and brightness)	—

The texture is saved with the project, and each song section (A–D) has its own. The generated sources use no memory; at SRC OFF nothing runs.

Per track

- **FILTER** (FX → FILTER, or **FX + KNOB 4** for the selected track): one knob, left a low-pass, right a high-pass, centre off. It stays when you change the sound, can be locked on a step, and is saved with the project. Its **resonance** is set once for every track in HOME menu → AUDIO → **FILTER RESO** (0 clean … 10, 2.7.2). **(2.8) The drum track has it too:** FX → FILTER sweeps and shapes the whole kit, live.
- **Sends** (FX page): DST drive, CHO chorus, DLY delay, REV reverb. The delay and the reverb / chorus settings are global (DLY and REV/CHO pages). The delay's TIME includes dotted 1/8 and 1/16.
- **SLICER** (FX → SLICER): a tempo gate (GATE) or stutter (STUT), 16 patterns, 1/8 to 32T, with a depth.
- **INSERT** (FX → INSERT and INSERT 2, 2.6, after Felucca 1.5): one effect on the track alone, right after its drive; the drum track has one too, on the whole kit and its sends. MIX sets dry / wet. A, B and C change

with the type:

Type	A	B	C
SOFT (soft clip), **HARD** (hard clip), **FOLD** (wavefolder), **FUZZ**	DRIVE	TONE	LEVEL
CRUSH	BITS 1–16	RATE (the sample rate, 689 Hz – 44 kHz)	LPF
PHASR (phaser), **FLANG** (flanger), **CHOR** (chorus)	RATE	DEPTH	FDBK feedback

A, B, C and MIX can be locked on a step. TYPE OFF (or MIX at 0) costs nothing: the track sounds exactly as without it.

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19. The visualiser

On the TRACKS screen, **tap HOME**: the whole screen becomes a visualiser of what SLOOP plays.

Control	Does
SELECT	the next / previous style (its name shows a second; the last one is kept with the settings)
HOME, or any page button	closes it
keys, PLAY, REC, layers	work as ever (a held layer shows its screen, then the visualiser comes back)
KNOB 1–4	nothing (the tempo is GLO + SELECT meanwhile)

#	Style	Shows
1	**OSCILLOSCOPE**	the mix's wave, standing still
2	**SPECTRUM**	32 bands with falling caps
3	**SPECTROGRAM**	the spectrum scrolling down, black → blue → green → yellow → white
4	**LISSAJOUS**	the stereo image: the wider the cloud, the wider the sound
5	**VU METERS**	tracks 1–4 in their colours and the mix, with peak holds
6	**CIRCLE**	the wave round a ring that swells on every kick
7	**TAPE**	two reels turning with the song, the level, the bar and the tempo
8	**LCD**	the tempo in seven-segment digits, bar and beat, four track meters
9	**BOUNCE**	a ball per track, kicked up by its notes
10	**ORBIT**	four planets turning in 1, 2, 4 and 8 beats round a sun that pulses with the mix
11	**WIRES**	a string per track, set swinging by its notes
12	**SLOOP**	the logo alive: the sail's four bands are the tracks, the boat rocks on its wave

The visualiser sees the mix **as if MASTER were all the way up**: with MASTER turned down, even to 0, it still moves at full size. It costs the sound nothing.

20. The HOME menu and the lights

Hold HOME (0.7 s) to open the menu; hold it again to close it. The menu is in five sections (2.7.1: SEQ):

Section	KNOB 1	KNOB 2	KNOB 3	KNOB 4
SCREEN	**COLOR**: the screen palette (GREEN, AMBER, CYAN, RED, MONO, then your theme once you sent one)	**ZOOM**: ON shows the value you touch large and white	**KNOB ACCEL** (2.7): ON (the default) a fast turn sweeps wider; OFF one value a detent	—
LIGHTS	**LIGHTS**: OFF, LOW, MID, HIGH	**KEYS**: OFF, C KEYS, WHITE KEYS, ALL KEYS	**NOTES**: OFF / ON	—
AUDIO	**SPEAKER LOWCUT**: ON cuts the lows the small speaker cannot play (~110 Hz)	**USB AUDIO**: MASTER / FULL	**USB RATE** (2.7): AUTO / 44.1 KHZ / 48 KHZ (takes effect at the next start)	**FILTER RESO** (2.7.2): the resonance of the per-track filter, 0 (clean, as before) … 10
SEQ (2.7.1)	**STEP ADVANCE**: ON (the default) the NOTES page moves on when you let go of the keys; OFF it stays, so a chord you missed is played again on the same step (KNOB 1 moves the cursor)	**DRUM PREVIEW**: OFF (the default) a drum sound picked with a knob (DRUMS screen, SEQ layer) is not heard, quiet for live; ON you hear it, as up to 2.7	—	—
SYSTEM	**2ND CORE** (2.7): **ON** (the default) / OFF. ON, the sound is computed on both of the FM-1's cores: about twice the room before the CPU runs out (many voices, INSERT effects on several tracks), the same sound. ON takes effect at the next power-on; ABOUT then says *2 CORES* (*1 CORE*: the second core did not answer and SLOOP runs on one, as before). OFF: one core, as in 2.6	**HARDWARE CALIBRATION** (the knob selects it, OCT+ opens it)	**MIDI LEARN CLEAR** (2.6): how many CCs are learned; OCT+ forgets them all	**ABOUT**, the version (the knob selects it, OCT+ opens it)

The knobs (2.7). A slow turn is fine tuning, one value a detent. Turn faster and the value sweeps wider, progressively (x2 up to x16 on the widest values); turning back to correct is always one value a detent, and lists of names never jump. MENU → SCREEN → KNOB ACCEL OFF keeps one value a detent everywhere. **LEVEL and PAN glide** over about 3 ms instead of stepping at each detent, on every track and the drums: no zipper on a held note. Both after JIANT FM by juanjiant-bit.

Control in the menu	Does
SELECT	the previous / next section
KNOB 1, 2, 3	set the section's rows directly (each row shows its knob's colour)
PRESETS	moves the cursor
OCT+	steps the cursor's setting round, or opens it (CALIBRATION, ABOUT)
OCT–	closes (from ABOUT: back to the section)
hold HOME	leaves the menu

All of these are **settings of the FM-1**, kept when you load a project or start a new one.

Lights

- **LIGHTS** — every button glows at that level, so its label can be read in the dark (on a black FM-1 the labels are unreadable unlit). What is on — the page, PLAY, REC, an octave — stays at full light and still blinks.
- **KEYS** — the Cs, every white key, or every key glow at the LIGHTS level too (KEYS turns LIGHTS on at LOW if it was off).
- **NOTES** — ON: the notes sounding on the selected synth track light their keys, played live or by the sequencer (each for at least a tenth of a second, so short sequencer notes show too). The drum track always lights its hits. Works on every page and in every layer.
- **Always:** PLAY flashes on every beat; REC blinks when armed and is solid while recording; the button of the page shown is lit; in a layer, keys 1, 5, 9, 13 glow dimly and what is on is fully lit; OCT– / OCT+ light when the octave is off 0; a locked layer's button blinks.

Hardware calibration

If a button or knob does the wrong thing, HOME menu → SYSTEM → **HARDWARE CALIBRATION** (or hold **OCT– + OCT+** while switching on): press each button and turn each knob to the right as the screen asks. 30 s without input cancels it and keeps the old setup.

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21. MIDI

Where MIDI comes in

- **The MIDI IN jack** (3.5 mm TRS on the FM-1): a keyboard or pad controller with a MIDI output, through a TRS-to-DIN MIDI adapter. If nothing plays, try the other type of adapter (type A / type B).
- **USB**: a computer or phone (a DAW, a MIDI routing app) or a USB MIDI host box. A USB keyboard plugged straight into the FM-1 cannot work: both are USB devices, and a USB link needs a host.

Both work at once.

All the MIDI settings: GLO → MIDI, MIDI 2 (2.6) and MIDI 3 (2.7)

Every MIDI setting is on these pages of GLO (tap GLO until **MIDI**). They are settings of the FM-1: kept through power-off, not changed by a project, and in the web editor's backups.

Page	KNOB 1	KNOB 2	KNOB 3	KNOB 4
MIDI: which channel each track uses, in and out	**CH1** track 1 (default 1)	**CH2** track 2 (2)	**CH3** track 3 (3)	**CH4** the drum track (10)
MIDI 2: how MIDI comes in and goes out	**IN**	**OTHER**	**OUT**	**BEND**
MIDI 3 (2.7): the punch-in effects, the drums, the buttons over MIDI, the clock out	**FX CH**	**DR MAP**	**REMOTE**	**CLOCK**

- **CH1–CH4:** OFF or 1–16. A track's channel plays it (notes, CCs, sustain, pitch bend, Program Change) and its keys and sequence go out on it. **OFF**: nothing in, nothing out for that track. Two tracks never share a channel: turn a channel onto one another track has and it goes on to the next free one, and the screen says whose it was (*CH 3 = TRACK 3*). Changing a channel ends whatever was still sounding on it, here and on the other gear. CH4 is also on GLO → DRUMS (the same setting).
- **IN: NOTES** (notes, CCs and the clock: normal) · **CLOCK** (only the clock and START / CONTINUE / STOP; every note and CC ignored, except the REMOTE CCs when REMOTE is on) · **CH 1 … CH 16** (only that one channel, always on the selected track; CH1–CH4 and OTHER are set aside, the clock still followed). Opening GLO → MIDI while IN is not NOTES says so on the screen (*IN CH 5: SEL TRK ONLY*, *IN = CLOCK: NO NOTES*).
- **OTHER:** what a channel that no track has does: **SEL** plays the selected track (set your keyboard to channel 4 and it follows ALGORITHM) · **OFF** ignores it (a sequencer's channels for other gear on the same cable).
- **OUT: KEYS** only what you play on the keys goes out · **SEQ** the sequencer, the arp and the rolls go out too.
- **FX CH** (2.7): OFF (the default) or 1–16. On that channel the notes **F3 … G5** (MIDI 53–79) are the FM-1's 27 keys with **FX** held: a note held holds its punch-in effect, its release lets it go (white keys: effects 1–16, black keys: 17–27, as on the FM-1). A pad controller, a keyboard or a DAW can play and automate them. That channel only plays effects (choose one no track uses); it works with IN = NOTES or CH n, not with IN = CLOCK. A key of the FM-1 (FX + key) takes over at once.
- **DR MAP** (2.7): how the notes reaching the drum track (on CH4, or on IN CH n with the drum track selected) pick the 16 drum sounds. **GM** (the default, as before): the General MIDI notes, for DAWs and MIDI files: 36 kick, 35 kick 2, 38 snare, 39 clap, 42 hat, 46 open hat, 44 pedal hat, 37 rim, 40 snare 2, 43 low tom, 48 high tom, 49 crash, 51 ride, 70 shaker, 63 conga, 56 cowbell (any other note plays the nearest of them). **1-16**: any 16 notes in a row play the 16 sounds in order, from note 36 (KICK, KICK 2, SNARE, CLAP, HAT, OPEN HAT, PEDAL, RIM, SNARE 2, LOW TOM, HI TOM, CRASH, RIDE, SHAKER, CONGA,

COWBELL), and again every 16 notes: a keyboard or a row of pads plays the whole kit wherever it starts. What the drum track sends on MIDI OUT stays GM.
- **REMOTE** (2.7): OFF (the default) or 1–16. On that channel, **CC 102 … 115** press the FM-1's 14 buttons from a controller or a DAW: a value of 64 or more holds the button down, below 64 lets it go (a controller's button sends 127 then 0: a tap; held: a hold). Everything works as on the panel: taps, holds and their layers, combinations (a button held + a key), OCT– / OCT+. CC 102 FX · 103 SEL · 104 ENV · 105 LFO · 106 EDIT · 107 GLO · 108 HOME · 109 SAVE · 110 ARP · 111 SEQ · 112 PLAY · 113 REC · 114 OCT– · 115 OCT+. Choose a channel no track uses; its other messages work as usual (notes, other CCs, MIDI LEARN).
- **CLOCK** (2.7): OFF (the default) or **OUT**. OUT, the FM-1 is the **clock master** on USB: it sends the MIDI clock (24 pulses a beat, at its tempo, also while stopped so the other gear knows the tempo), **START** when its sequencer starts (after a count-in, when it starts) and **STOP** when it stops. A DAW, a drum machine or another groovebox set to follow MIDI clock then plays in time with SLOOP and starts and stops with it. It works with SYNC = INT only: a FM-1 that follows a clock (SYNC USB or TRS) never sends one (no loop through a computer). The FM-1's TRS jack is an input: the clock goes out on USB. A setting of the FM-1, kept through power-off.
- **BEND:** the pitch bend's range, 1–12 semitones (default 2).

IN and OUT are also on GLO → SYSTEM, next to SYNC (the clock), where they were before 2.6.

Examples. A keyboard on channel 1 plays track 1; select another track and play it from channel 4 (OTHER = SEL). An external sequencer with drums on 10, bass on 2, pad on 11, lead on 8: CH1 = 2, CH2 = 11, CH3 = 8, CH4 = 10, OTHER = OFF. A KeyStep Pro whose other channels drive other gear: IN = CH n, and that channel plays the selected track.

MIDI out, clock in (GLO → SYSTEM)

Knob	Setting	Choices
KNOB 1	**OUT** (MIDI out)	as on MIDI 2
KNOB 2	**SYNC** (clock in)	**INT**: SLOOP's own tempo · **USB** or **TRS**: follow that MIDI clock
KNOB 3	**IN**	as on MIDI 2
KNOB 4	**CPU**	the processor load (or the USB link state while no computer is connected)

- **MIDI out:** each track on its channel (CH1–CH4). With **SEQ**, set a DAW track to record the FM-1's MIDI and you get the pattern as notes, or let SLOOP drive another synth. Every note is ended, STOP ends whatever was still on, and notes that came in are never sent back (no MIDI loop).
- **Clock in:** START plays from the top, CONTINUE carries on, STOP stops; the tempo follows the master pulse by pulse (24 a beat), so SLOOP cannot drift. When the clock stops for half a second, PLAY on the FM-1 plays at its own tempo again. A MIDI START during the REC count-in starts and records at once.
- Switching IN, or a channel, releases any note still held: nothing is left hanging.
- **Bluetooth MIDI** is not supported: SLOOP never switches the radio on.

MIDI CCs (2.5)

The knobs of a MIDI controller set the sound (the standard CC map of Felucca 1.1.5). A CC acts on the track its channel plays, as the notes do (each synth track's channel, the drum channel the drum track, the others the selected track), and sets its parameter as a knob would: 0–127 over the parameter's range, 64 the middle of PAN and FILTER. It changes the sound's value itself (as a knob, so saving keeps it); the screen and the web editor follow.

CC	Sets
5	GLIDE
7	LEVEL (on the drum channel: GLO → DRUMS → LVL)
10	PAN
11	(2.7.1) **EXPRESSION**: a volume pedal over the track's LEVEL (on the drum channel: over DRUMS LVL), 127 full, 0 silent, as on a GM synth. It does not change LEVEL (your mix and what you save stay as they are). Full again at power-on, when a project or section is loaded, and with CC 121
121	(2.7.1) RESET ALL CONTROLLERS: the expression full again
71	the engine's resonance (RES or Q: ANALOG, TRIO, VOICE, NOISE; the others ignore it)
72 · 73 · 75	release · attack · decay
74	(2.7) the engine's own cutoff or brightness, dark to bright: ANALOG / SAMPLE CUT, TRIO CUT, PHASE DCW, LOFI / GRAIN TONE, FM6 MLVL, PHYS BRIT, NOISE FREQ. DIGITAL, VOICE, WHEEL and the drum track: the track's FILTER (64 off, lower a low-pass, higher a high-pass), as before
91 · 93 · 94	the reverb, chorus and delay sends (on the drum channel 91 and 94 are GLO → DRUMS → REV and DLY)

Sustain and pitch bend (2.6). The **sustain pedal** (CC 64, on the track its channel plays): while it is down, a note you let go keeps sounding until the pedal comes up (the arp keeps it, a recording holds it as you hear it; a note played again meanwhile ends with its key). **Pitch bend** bends the track its channel plays, smoothly, by up to **BEND** semitones (GLO → MIDI 2: 1–12, default 2); the drum track ignores it. Other CCs (mod wheel…) are ignored, unless you map them with MIDI LEARN.

Program Change (2.6, after Felucca 1.5.1 and @renebohne's PR #179). A Program Change on a track's channel loads its sound: on a synth track, number *n* is the *n*-th sound of the PRESETS list (0 the first: 808 BOOM; the 153 factory sounds by kind, then your user presets in order), on the drum channel kit *n*. A number past the end is ignored. The screen says *T2 PROG 5*. IN = CLOCK ignores it; IN = CH n sends it to the selected track. **(2.7.2) Bank Select (CC 0)** reaches the sounds past the first 128: send **CC 0** on the track's channel and it is latched for the next Program Change — bank 0 the first 128 of the PRESETS list, bank 1 the next 128, and so on (the rest of the factory sounds, then your user presets by slot). Send CC 0, then the Program Change.

MIDI LEARN (2.6)

Any knob of your controller onto any parameter of a track (after Felucca 1.5):

1. **GLO + F#3** (the first black key): *LEARN: TURN A KNOB*; the top line shows **LEARN**.
2. Turn the knob of SLOOP you want (any page of the track, or LEVEL / PAN on TRACKS): *T1 CUT: SEND A CC*.
3. Move the knob on your controller: *CC 21 = T1 CUT*. Go on with other knobs, as many as you like.
4. **GLO + F#3** again: *LEARN DONE*.

A learned CC sets that track's parameter whatever channel it comes on, and no longer does what the standard map gave it (a CC learned for one parameter is taken off the one it had before; up to 16 CCs). The CCs with a meaning of their own are never learned: bank select (0, 32), mod wheel (1), data entry (6, 38), expression (11), sustain (64), 96–101 and the channel modes (120–127). What is learned is a setting of the FM-1, kept through power-off and projects, and in the web editor's backups (restored with them, on this FM-1 or another); **HOME menu → SYSTEM → MIDI LEARN CLEAR**, OCT+, forgets it all.

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22. USB audio

On USB the FM-1 is also an audio input named **Felucca**: 44.1 or 48 kHz (the computer, phone or tablet picks; 48 kHz since 2.5, for the phones and apps that only take it), 16-bit stereo, class compliant, no driver. In your DAW, in Audacity or in a sampler app on your phone (with a USB-C cable or adapter) choose that input and record: you get the master output, exactly what the headphones play (after DUST, DUCK and FILT; the click and count-in too if they are on). MIDI, the editor and the installer keep working on the same cable.

HOME menu → AUDIO → USB AUDIO:

- **MASTER** (default): the recording follows the MASTER knob, as the headphones do. Keep MASTER well up while you record.
- **FULL**: a fixed level, as with MASTER all the way up (kept from clipping by the limiter); MASTER then only sets the headphones. Best for a computer input with no level control of its own.

The first time, the computer sees the FM-1 as a slightly different device (MIDI + audio) and sets it up again. On macOS 13–15, keep **USB SERIAL** OFF (the default) or the audio input may not show (2.6; from 2.7 the FM-1's build has no serial console and no USB SERIAL: the flash is full, and it was a developer's tool). (2.7.2) On macOS 15 and older, SLOOP up to 2.7.1 showed MIDI but no audio input: Apple's audio driver there stopped reading SLOOP's USB description before the audio part (after Felucca 1.5.1, the audio is now listed first). The FM-1 also reports a new USB version, so the Mac sets it up fresh.

HOME menu → AUDIO → USB RATE (2.7), KNOB 3: the rates the FM-1 offers. It takes effect at the next start: turn the FM-1 off and on (the menu shows *RESTART* until then).

- **AUTO** (default): 44.1 and 48 kHz, the computer or phone picks, as in 2.5 and 2.6.
- **44.1 KHZ**: 44.1 kHz only, the samples exactly as the FM-1 makes them (as in 2.4). If a phone or tablet records the FM-1 distorted or noisy while a computer records it clean, try this first: the phone then converts the rate itself.
- **48 KHZ**: 48 kHz only, for a device or app that needs it.

The computer or phone sees each setting as a slightly different device and may set it up again.

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23. Your own samples and drum kits

The FM-1 has **four sample slots**, **USR1–USR4**, of about **7.4 s** each (22 kHz mono). A slot holds either:

- **an instrument**, played by a synth track: engine **SAMPLE**, **SET = USR1 … USR4** (or the GRAIN engine's SRC), or
- **a drum kit**, played by the drum track: **KIT = USR1 … USR4**, or **USR3+4**, one big kit of about **15 s** over two slots.

A good setup: USR1 and USR2 as instruments, a big kit on USR3+4. You fill the slots from the web editor.

An instrument (editor → Samples)

1. **Choose a slot.** Four tiles show what each slot holds and how full it is; the page warns when sending would replace something (or half of your USR3+4 kit).
2. **Make the sound**, one of two ways: - **From files (one note per file):** drop up to 16 WAV files (any rate, mono or stereo). Each plays at its own note, the keys between play the nearest one, pitched. The note is read from the file name (`KEYS_C4.wav`, C4 = 60) or set in the list; a keyboard picture shows which keys play which file; ▶ plays a file; the meter shows the time used. Each file has a **Gain** (0 to –24 dB, 2.6); **Match levels** sets them so every file is heard at the same level (the loud ones turned down: every file comes in at full scale), then adjust by ear. **Loop** (2.7), for each file: **ONE SHOT** (plays once to the end, as before), **LOOP** (the whole file again and again while the key is held; let go, it ends its round) or **LOOP+TAIL** (the start plays, then the part between **Loop start** and **Loop end** again and again while the key is held; let go, the rest of the file, the tail, plays to the end). A short sample becomes a sound you can hold: a pad, a string, a voice, an organ. Set the points on the file's wave (the loop is lit); when you let go of a slider the point moves to the nearest zero crossing, so the loop does not click. Hold ▶ to hear it as the FM-1 plays it, let go for the tail. A WAV that has its own loop (set in a sample editor: its *smpl* chunk) comes in as LOOP+TAIL with its points. It takes no extra memory. The SAMPLE engine's **LOOP** (EDIT 1, KNOB 4) still turns loops off for the whole set. The release (ENV) still ends the note: set REL long enough to hear the tail. On SLOOP 2.6 such a file loops until the release ends it. - **Chop a recording** (CHOP): open or drop a recording (WAV, MP3, AIFF…) and cut it into up to 16 chops, one per key.
3. **Send it and play it:** name it, **Send to USRn**, then **Play this slot on track 1, 2 or 3**: that track switches to SAMPLE with SET = USRn.

CHOP in detail:

Control	Does
TAP (or the space bar) while it plays	a cut at that moment (*snap to the hit* puts it on the attack)
Find hits (with a sensitivity), **Grid**, **Equal parts**	cuts made for you
the box on a chop, **Keep all**, **Keep none**, **K**	keep or leave out chops (left-out chops go neither to the slot nor to the WAVs)
the length slider, or the handle at the bottom of the wave	shorten a chop; **Up to the next marker** undoes it
Fit to slot	shortens the longest chops just enough to fit
Download WAVs	the kept chops as WAV files
keys 1–9, 0	listen to a chop
double-click / drag / ← → / Del	add / move / nudge / remove a marker
mouse wheel	zoom

A recording of any length works: keep the chops you want, the slot takes about 7.4 s of them.

A drum kit (editor → Drum kit)

1. **Your sounds.** Drop a WAV on any of the **16 pads** (KICK, KICK 2, SNARE, CLAP, HAT, OPEN HAT, PEDAL, RIM, SNARE 2, LOW TOM, HI TOM, CRASH, RIDE, SHAKER, CONGA, COWBELL), or **Choose files** to add several at once: they are sorted onto the pads by name (*kick*, *bd*, *snare*, *hh*, *open*, *crash*…; a second kick goes to KICK 2). Click a pad's wave to hear it. Each pad has **Pitch** (±12 semitones), **Gain** and **Length** (a cut ends with a short fade). **Match levels** (2.6, under the pads) sets every pad's Gain so the kit is heard even (as the ear hears it: the lows count less; the loud pads turned down to the quietest), then adjust by ear.
2. **Where it goes. USR3+4** (the big kit, about 15 s; the editor shares the sounds out over the two slots by itself), or a single slot **USR1–USR4** (about 7.4 s). Each choice shows what it holds now and what a send would replace. The meter shows the time used; too long? shorten the long sounds or press **Fit the slot**.
3. **Send it.** Name it, **Send**, then **Use … on the drum track** selects track 4 and sets its KIT (or turn PRESETS on the drum track: USR3+4 is the last kit).

Save as ZIP keeps the kit's 16 WAVs to share; **Open ZIP** (or dropping a .zip) loads one back.

On the FM-1 your kit works like any other: steps, levels, ratchets, fill conditions, the DRUMS grid, the keys, MIDI channel 10 (a note plays its lane's sound), the filter, the slicer, mute. The closed hat cuts the open one. A lane without a sound is silent, and so is an empty slot.

Your own synthesised kits (SYN1–SYN4)

SLOOP 2.5: the editor's **Drum synth** page changes every value of a synthesised kit, sound by sound, and the FM-1 plays the change at once. Never stored, SYN1–SYN4 are copies of 808, 909, TRAP and TECHNO.

1. **Pick a kit**, SYN1 to SYN4, and give it a name (8 letters: the DRUMS screen shows it). **Start from** + **Copy** makes it a copy of any of the 32 kits.
2. **Pick a sound** (KICK, SNARE, CLAP, HAT, OPEN HAT, LOW TOM, HI TOM, CRASH, RIDE, SHAKER, CONGA, RIM, COWBELL, CLAVE, KICK 2, SNARE 2): it plays (also **Play**; **Play on change** plays it after every change).
3. **Change it.** Each value is a tile you drag up or down, or a bar; **double-click** puts back the value of the kit it started from.

Group	Values
Tone	**WAVE** (OFF, SINE, TRI, SQUARE, FM, BELL) · **PITCH** (a note, its Hz) and **FINE** · **BEND** (semitones the pitch falls from at the hit) and **B.TIME** (how fast) · **HOLD** (at full level) and **DECAY** · **LEVEL** · **2ND** (a second partial: its ratio, the drum head's second mode) and **2ND LVL** · **CLICK** (the attack: the beater, the stick)
Noise	**NOISE** (OFF, WHITE, METAL: the 808's six squares, CYM: metal and white, CHIP: an 8-bit noise) · **CLAP** (three bursts, then the tail) · **LEVEL**, **HOLD**, **DECAY** · **HPF** (a high-pass on the noise) · **CHIP** (the chip noise's clock)
Filter	**FILTER** (OFF, LP, BP, HP) · **ON** (the noise, or ALL: the tone too) · **CUTOFF**, **RES** · **ENV** (octaves the cutoff starts above it, falling with the pitch)
Out	**DRIVE** · **LEVEL** (dB)

CRUSH (bits off) and **S&H** (a lower sample rate) act on the whole kit.

4. **Store on the FM-1** keeps the four kits in flash (stop the song first); until then *not stored yet* shows (switched off before, the FM-1 may come back with the last stored kits). **Use on the drum track** sets KIT to it. **Save file / Open file**: a kit as a file, to share or to keep. A backup holds the four kits too.

24. The web editor

Open it from the installer page, the editor link, or `OPEN-EDITOR.bat`, in **Chrome or Edge** with the FM-1 on USB, and press **Connect** (allow MIDI). It follows the device live: turn a knob on the FM-1, the editor moves. It looks like the FM-1: the black screen, the four track colours, the FM-1's own pixel font. The language button switches English / Japanese.

Offline (2.6). *Download the editor* (under Connect, or on the installer page) saves the whole editor as one file for a computer, `sloop-editor-2.6.html`, also attached to each release. Double-click it: it opens in Chrome or Edge without internet (Chrome asks once to allow MIDI) and works as the online one. Its sound library (kept in the browser) is its own, apart from the online editor's: move sounds between them with Export / Import, or a backup.

As an app, also on Android (2.6). Open the online editor or the installer once with internet, then *Install the app* (under Connect, or on the installer page; or Chrome's menu → *Install app* / *Add to Home screen*). From then on the SLOOP app opens the editor without internet, and its link to the installer works offline too (with the firmware it installs). It stays the online site kept by the browser, so it uses the same sound library as the online editor, and it takes each new SLOOP version by itself the next time it opens online. On Android, use the app rather than the downloaded file: Chrome on Android does not allow MIDI in a downloaded file.

Layout: the side bar has the pages in three groups — **Track** (LIVE, Sound, Sequencer, Tracks, Song), **Sounds** (Library, Samples, Drum kit, Drum synth), **Device** (Projects, Settings) — and the connection. On top: the page title, the messages, and on the Track pages the **four tracks** (number, sound, engine): click one to select it. With the side bar focused, the arrow keys move between pages.

Values: each value is a tile you **drag up or down like a knob** (Shift: fine), or a bar: click it, then the wheel or the arrow keys (Shift: steps of 10); **double-click** resets it.

Page	What you do there
LIVE (2.7)	the whole FM-1 under your hands for a show, from a computer, a tablet or a phone: see LIVE
Sound	every parameter of the selected track, as on the device: a card per button (ENV, LFO, EDIT, VOICE, FX, SEL, ARP), its pages as rows of four coloured values. **Engine** and **Presets** pickers, **Init sound**, **Copy to library**, **Save to file / Load file** (a sound as a file). On an FM6 track, the **FM6** panel: the six operators, **Read from track**, **Send to track** (*Send while editing* to hear every change), **Store in bank** (B1–B27), **Import SysEx** (or drop a .syx), **Store cartridge in bank** (2.5: the ticked voices of a bank into B1… in one write, and as user presets if you choose), **Export SysEx**, **Export bank as SysEx**. **The drum track** (2.7): **DRUMS** — the kit (‹ ›), level, pan, filter, reverb, delay, the INSERT, and the 16 pads: tap one to hear it, its **tune** and **decay** as the DRUMS screen's SOUND page (**all pads as the kit** puts them back)
Sequencer	the pattern settings (LEN, DIV, SWG, GATE) and the steps as the device's tiles; the **step list** takes typed note names (*C4 E4 G4*). On the drum track: a **grid** of the 16 sounds × the steps, with the **kit**: choose a **level** (GHOST, SOFT, NORM, HARD) and a **roll** (x1–x4), then click a cell — a hit; click again (same level and roll) — cleared; **Shift+click** — one level louder. Click a step for its **detail**: **Nudge**, **Parameter locks** (+ lock, pick the parameter, set or delete), **Fill** (normal, fill only, no fill). **Reload**, **Clear sequence**. **Length**: – / + or type 1–64. **Grid / Piano roll** and **Export MIDI / Import MIDI**: see below
Tracks	the four channel strips: level, pan, mute, select; SOLO and REC shown as on the device
Song	the song order of the SONG screen: a row per section (A–D), its **bars** (1–64) and how many **times** in a row; ▲ ▼ move a row, ✕ removes it, **Add**; **At the end**: stop or start again. **Read from the FM-1**, **Send to the FM-1** (16 entries in all; stop the song first). A section whose slot is empty is marked: save it on the FM-1 first (SAVE + key 5–8)

Page	What you do there
Library	your sound library, kept in the browser: **Save current sound**, search by name or tag, filter by engine, sort, **Audition**, **To device slot**, **Rename**, **Tags**, **Duplicate**, **Delete**, **Export** / **Import** / **Export library**; copy factory presets in. Next to it the **32 user presets on the device**: **Load**, **Store current sound**, **To library**, **Erase**, **Export bank**. Drag sounds between the library and the slots
Samples	your instruments in USR1–USR4, in three steps (see above), with CHOP
Drum kit	your own drum kits, in three steps (see above)
Drum synth	your own synthesised kits SYN1–SYN4: every value of every sound (see above)
Song & sections	the sections A–D (**Load** into the working loop, **Save** it into one), **Songs** (2.7: **Export this song**, the loop, A–D and their order as one *.sloopsong* file; **Load a song file**; the LIVE page's playlist plays them) and **Backup** (everything on the FM-1)
Settings	GLOBAL (BPM, SWING, CLICK, TUNE), **MASTER** (DUST, DUCK, FILT, ROLL), DRUMS (CH, LVL, REV, DLY), and the system information (firmware, sync, slots)
Theme (2.7)	your own screen colours: see Your theme

LIVE (2.7). The editor's **LIVE** page plays the FM-1 in front of an audience: everything a show needs on one screen, with the mouse, a finger on a tablet or a phone, or the computer's keys. What its buttons do goes through the FM-1's own code (the same as GLO + key, SAVE + key, FX + key, REC, PLAY), so the FM-1 and the page always agree, and you can play on both at once.

- **Transport**: **PLAY / STOP**, **REC** (the selected track, as the REC button), **undo / redo**, **BPM** (drag the knob, or **TAP**), **SWING**, where the loop is (bar.beat) and, in song mode, which part and bar of the song.
- **Song**: **LOOP / SONG**, **SONG REC**, **FILL** while held and **FILL BAR** (the next bar is a fill), as GLO + key 9 and 10.
- **Sections A–D**: tap one: it plays from the next bar (stopped: at once); the one playing is lit, the one coming blinks, an empty one is dim. **Save the loop into A B C D** stores the loop (over a used section it asks first).
- **Mixer**, one strip per track: the **sound** (any preset; the drums: the kit) with ‹ › for the next one, the **level** fader with its meter, the pattern's progress, **filter** (left: low-pass, right: high-pass), **pan**, **reverb**, **delay**, **chorus** and **drive** (the drums: filter, pan, reverb, delay), **mute**, **solo**, and **play**: the keys below play that track (it is selected on the FM-1 too, so REC records it).
- **Master FX**: the master **filter**, **dust** and **duck** (FX + KNOB 1 to 3 on the FM-1).
- **Punch-in effects**: the 27 effects in four groups (loops & time, filter & EQ, colour, movement). Hold a pad: the effect plays while you hold it, as FX + key. **Latch**: a tap turns it on, another tap (or another effect) off.
- **Keys**: two octaves for tracks 1–3 (octave, velocity, the track's **arp** mode and **hold**), the 16 drum pads for the drum track (GHOST, SOFT, NORM, HARD). They are real MIDI notes on the track's channel (GLO > MIDI), so they sound at once.
- **Playlist**: your songs, one after the other. Export each song first (**Song & sections** → **Export this song**: the loop, the sections A–D and their order in one *.sloopsong* file), **add songs** here, and tap one (or **next song**, key **N**): the FM-1 stops, the song loads in a few seconds (its A–D, loop and order replace the FM-1's) and, with **play when loaded**, starts. The list stays in the browser; **save the list** keeps it as one file to take to a gig (**add songs** opens it again).
- **Full screen**: the page alone, the side bar hidden. **All off** lets go of every note and effect.
- **Help on the page**: every card says in one line what it is for, its **?** explains how to use it, and the mouse over any control says what it does. **First live in 5 steps** shows at the top until you press *got it* (its **?** brings it back). **view: essentials** keeps the transport, sections, mixer, keys and effects; **view: everything** adds the scenes, PERFORM, the playlist and the stage tools.
- **Sections or scenes?** The sections A–D change **what plays** (the notes: intro, verse, chorus…). The scenes change **how it sounds** (the knobs: filters, levels, effects), never the notes. Example: scene 1

"intro" with the filters closed and the drums low, scene 2 "drop" with everything open: play your loop, tap 1, then 2 at the drop — or let scene 2 come over 8 bars for a build-up.

- **Scenes** (8): a scene keeps how everything sounds now — each track's level, mute, pan, filter, sends and drive, its sound's knobs (the engine's 8 values, FLT PIT SHP, the envelope, glide, the LFO), its INSERT and arp, and the master bus (filter, dust, duck, the delay, the reverb, the chorus, TEXTURE, swing). **+ sounds** keeps the presets and the kit too, **+ tempo** the BPM. **Store** then a scene pad keeps it (double-click: its name); a tap recalls it **at once** or **morphs** there over 1 to 16 bars. The **crossfader** mixes two scenes (A and B) as you move it. Kept in the browser, and in the setlist file.
- **Perform**: the track the keys play, in four tabs: **sound** (the engine's values, the macros, the envelope and glide, the LFO), **arp & key** (MODE, RATE, OCT, GATE, SWG, ORD, HOLD; ROOT, SCALE, CHORD, TRANSPOSE, for **every synth track** at once if you want), **insert & bus** (the track's INSERT and filter; the delay, reverb, chorus, TEXTURE and the metronome) and **steps** (the 16 steps of a page, a drum lane a row: a tap turns a step on or off, a new synth step plays the last note you played; a right-click its fill condition; LEN, DIV, SWG, GATE).
- **Sounds**: ☆ on a strip makes its sound a favourite (at the top of every list, in this browser); your **USER** presets are in the lists too.
- **MIDI learn**: press **MIDI learn**, tap a control of the page (a fader, a knob, a scene, an effect pad, PLAY…), then move a knob or press a pad on your controller (any USB MIDI controller plugged into the computer). Learned controls work as soon as the page shows; a right-click in learn mode forgets one, **forget all** every one.
- **The FM-1's live tools** (2.7): **chain** (under the sections: tap 2+ letters, **play chain**: each section for its bars, looped, as SAVE + several keys), **roll / erase** for the keys and pads (roll repeats while held at the **roll rate**, as ARP + key; erase removes that note or sound while held, as EDIT + key), the pattern's **shift ◀ ▶, ½ / ×2 length, transpose** (PERFORM → steps, as EDIT + KNOB 1–3), each **pad's tune and decay** (PERFORM → sound on the drums), **rev throw / echo throw** while held and the **volume** (MASTER FX; turning the FM-1's MASTER knob, or **knob**, gives it back).
- **Stage tools**: the **clock** and the **set's time** (from the first PLAY; tap it to start again), **lock** (the page ignores taps, so nothing changes by accident; the keys of the computer and the controller still work; hold it to unlock) and **record set**: the FM-1's USB audio (the browser asks for the microphone once) recorded as a WAV file until you stop. In the playlist, ✎ gives a song its **tempo** (set when it loads) and a **cue** (shown while it plays, the next one under it).

The computer's keys (not in a text field): **Space** play / stop · **1–4** sections A–D · **5–8** mute tracks 1–4 (**Shift**: solo) · **9** fill while held · **0** fill bar · **B** tap · **C** REC · **N** next song · **Z / X** octave · ← → the track the keys play · **A W S E D F T G Y H U J K O L P** play (the drums: the 16 pads) · **Shift** + those keys: the first 16 effects while held · **F1–F8** scenes (**Shift**: store) · **Esc** all off.

The mixer and the keys also work with an FM-1 on SLOOP 2.6; the transport, sections, fills and effects need 2.7. If the keys stay silent, the page says why: GLO > SYSTEM > IN = CLOCK (no notes), or a track with no channel while OTHER is OFF.

Your theme (2.7). The editor's **Theme** page makes the FM-1's screen your own, with or without the FM-1 connected:

- **Screen**: five colours, faintest first (darkest on a dark background, lightest on a light one): **Lines**, **Dim** (empty steps, units), **Labels**, **Secondary** text, **Values**. **Start from** takes a built-in palette (the other way round on a light background); **Five steps from one colour** makes the five from the colour you pick.
- **Background**: black by default, or any colour, white too (**Back to black** returns to it). **Light ↔ dark** turns the theme over: a white background, the five steps the other way, a black accent (and back). The greys of the pages (tiles, empty steps, labels) follow the background.
- **Accent**: what you touch, the menu's cursor, the play head: white by default; on a light background a dark one.

- **Track colours**: T1, T2, T3 and the drums (the knobs' values, the step tiles, the meters).
- **Name**: up to 8 characters; the FM-1 shows it in MENU → COLOR.
- The preview draws a SLOOP page with your colours, exactly as the screen will show them.
- **Send to the FM-1** (stop the song first): the FM-1 keeps it and shows it at once. It then comes after MONO in **MENU** → **SCREEN** → **COLOR**; turn KNOB 1 to go back to a built-in palette and to your theme again. **Read from the FM-1** brings it back into the page; **Remove from the FM-1** deletes it. Send again after any change: next to the buttons the page says whether what you see is what the FM-1 has. **Values** and the **Accent** must stand out from the background so the menu stays readable: the page marks the one too close, Send says why, **Make it readable** moves just those two until they do, and the FM-1 refuses one too close. A theme made with a 2.7 beta still loads, with a white accent.
- **Share**: the line in the Share box (*SLOOP-THEME-1:…*) is the whole theme. Copy it into a comment or a message; anyone pastes it into the field below and presses **Use**.

A backup made with the editor keeps your theme. Back on SLOOP 2.6 the FM-1 shows the built-in palette you used last, and forgets the theme when it next saves its settings.

Piano roll and MIDI files (Sequencer)

Piano roll (a synth track): the steps left to right, the notes bottom to top (**Octave up / down** scroll). Choose a **note length** (1/1 to 1/16 of a bar, in steps of the track's DIV), then:

- click an empty spot: a note of that length. On the first step of a chord: one more note of it (4 at most, one length for all, as on the device);
- click a note: it goes (the last note of a chord leaves rests);
- drag the right end of a note: shorter or longer (ties), up to the next chord.

Every change goes to the FM-1 at once, and the grid shows the same steps.

Export MIDI saves the selected track's pattern as a MIDI file (`sloop-track1.mid`…): the notes with their lengths, levels and accents, at the song's tempo; the drum track as GM drum notes on channel 10.

Import MIDI reads a MIDI file into the selected track, at its DIV. First a preview: the file's track to take, how many notes and steps, and what changes on the way: notes off the grid moved to the nearest step, more than 4 notes on a step (the loudest kept), a note cut short by the next one, notes past step 64. **Import** replaces the track's pattern (and, ticked, sets its length); **Cancel** leaves it as it was. On the drum track, the GM notes go to the lanes (36 kick, 38 snare, 42 hat…) and the velocity to the levels, ghost to hard.

The editor writes presets, the FM6 bank and backups to flash only while the song is stopped, as the FM-1 does (it asks you to stop first). Try it without hardware: add `?mock=1` to the editor's address. The protocol is documented in web/EDITOR_PROTOCOL.md.

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25. Backup, rescue, going back, troubleshooting

Backup and restore

Editor → **Projects** → **Backup**:

- **Save a backup** writes everything on the FM-1 to one file (`SL0OP-backup-DATE.json`): the music you are working on, the sections A–D, the 32 user presets, the FM6 bank, the SYN drum kits, the samples **USR1–USR4** (your drum kits with them) and the settings (colours, calibration, song order, lights, SYNC, MIDI OUT).
- **Restore from a file** puts it all back: what is on the FM-1 is replaced. Stop the song (PLAY) first. A damaged file is refused before anything is written, and each object is written as a save writes it (a cut-off restore never leaves half an object).

A 2.4 backup restores into 2.4 (or later) only. **Save a backup before every update and before going back.**

Rescue and special modes

Situation	Do
SLOOP does not start	hold **OCT–** alone while switching on (*SLOOP USB RESCUE*), then install again from the browser
An install stopped half-way	the FM-1 waits in update mode: press **INSTALL** again
Buttons or knobs do the wrong thing	hold **OCT– + OCT+** while switching on, or HOME menu → SYSTEM → HARDWARE CALIBRATION
A developer needs UBOOT	hold **OCT– + OCT+** for 5 s while stopped (*UPDATE MODE IN 3…*; let go to cancel)

Going back to the official firmware

(2.8) The in-browser return-to-stock was removed, because it was not reliable on every system. To go back to the official firmware, use **M-VAVE's own updater, M-UPGRADE** (m-vave.com → Downloads → PC Software → M-UPGRADE) with the official **FM-1 V15** firmware (PC Firmware → FM-1 V15); close every other app that uses MIDI first. Save a backup with the editor before you go. To come back, install SLOOP again and restore your backup.

If the official firmware comes up with odd settings or a scrambled display, it is reading the settings SLOOP left in the FM-1's memory — nothing is damaged. A factory reset in M-VAVE's own firmware, or M-UPGRADE's full restore, clears it; and installing SLOOP again brings your work back exactly as it was.

Troubleshooting

Problem	Fix
The installer or editor does not find the FM-1	Chrome or Edge, a data cable, no hub, allow MIDI; close every other app or tab that uses MIDI, then reload
The black keys are silent on a synth track	it plays chords or the scale: SEL → CHORD OFF and KEYS OFF (with a chord on, the black keys change the chord: CHORD+)
Recorded notes move	SLOOP quantises to the track's DIV: REC screen QUANT off (2.7) keeps your timing; or use 1/32, or nudge the step (SEQ + step + KNOB 4)

Problem	Fix
Notes fade out on a dense part	the processor is at its limit; SLOOP fades one voice at a time (never the bass or lead) rather than glitching. Fewer held notes or a lighter engine help
Nothing from the MIDI IN jack	try the other adapter type (A / B); check the channel (GLO → MIDI: CH1–CH4; MIDI 2: OTHER) and that GLO → MIDI 2 → IN is NOTES
The USB audio input does not show	replug the FM-1; on a Mac (2.6) keep USB SERIAL OFF and restart the FM-1; in Audacity: Transport → Rescan Audio Devices
The USB recording is quiet	HOME menu → USB AUDIO = FULL, or MASTER up
The USB recording is distorted or noisy on a phone (clean on a computer)	HOME menu → AUDIO → USB RATE = 44.1 KHZ, then turn the FM-1 off and on
An FM6 sound changed after loading a project	store the edited patch in the bank (editor → FM6 → Store in bank) and set PTCH to it
STOP THE SONG FIRST / *STOP FIRST*	that change (new project, song edit, flash write) waits until the transport is stopped
EMPTY SECTION: REC	the section you asked for has nothing in it: save a loop into it first (SAVE + key 5–8)
Something else	open an issue: what you did, what you expected, what happened, and the version (HOME menu → SYSTEM → ABOUT)

26. Cheat sheet

Buttons alone

Do	Result
PLAY	start / stop
REC	record / arm / close a free take
hold REC ~2 s	clear the selected track
tap HOME	TRACKS · on TRACKS: visualiser
hold HOME	menu
tap SAVE	on TRACKS: SONG screen · else SAVE pages
tap ENV, LFO, FX, SEL, EDIT, ARP, SEQ, GLO	their pages (again: next page)
tap EDIT / SEQ on TRACKS, drum track	DRUMS screen (again: grid ↔ kit)
OCT– / OCT+	octave (both: 0) · drums: ghost / hard while held
ALGORITHM	track
PRESETS	sound · drum kit
SELECT	tempo · pages · visualiser style · menu section

Layers

Hold +	Keys	KNOB 1	KNOB 2	KNOB 3	KNOB 4	Other
FX	punch-in effects: 16 white keys, 11 black keys	master FILTER	DUST	DUCK	track FILTER	+ HOME: lock
EDIT	erase sound / note · black key: drum MUTE mode (white keys mute / unmute)	SHIFT	LENGTH ×2 / ½	TRANSPOSE	—	OCT– undo · OCT+ redo
ARP	note repeat	RATE	—	—	—	OCT– / OCT+: ghost / hard (drums)
SEQ	steps · black keys 1–4: page	SOUND / NOTE	DIV	SWING	LENGTH	OCT– / OCT+: page
SEQ + step	more steps	drums: which sound · synths: NOTE	drums: LEVEL · synths: VEL	RATCHET	NUDGE	PRESETS lock · ALGORITHM lock parameter · OCT+ fill condition · OCT– clear nudge, locks, condition
SEL	key of the song	CHORD	SCALE	KEYS	TRANSPOSE	—
GLO	1–4 mute · 5–8 solo · 9 fill · 10 fill bar · 16 tap tempo · F#3 MIDI LEARN	level 1	level 2	level 3	level 4	SELECT: tempo

Hold +	Keys	KNOB 1	KNOB 2	KNOB 3	KNOB 4	Other
SAVE	1–4 play A–D (several: chain) · 5–8 save A–D · 13 loop / song · 14 SONG REC · 16 song screen	—	—	—	—	—
any layer + HOME	lock it open					any other button unlocks

Chord mode (CHORD on)

Hold	+ white key
nothing	the scale's chord on that key (C4 = I)
F#	major ↔ minor
G#	+ 7th
A#	sus4
C#	+ 9th
D#	inversion

Power-on and special

Do	Result
OCT– held at power-on	USB rescue
OCT– + OCT+ held at power-on	hardware calibration
OCT– + OCT+ held 5 s, stopped	update mode (UBOOT)
EDIT + OCT– / OCT+	undo / redo
GO values (LOAD, SAVE, ERASE, NEW, CLRSQ, INIT)	one click arms (*AGAIN: …*), a second within ~1.5 s acts
SAVE + key 5–8 on a used section	press again within 3 s
SONG screen: REC on a used section · OCT+	press again within 3 s

Where is…?

Looking for	It is
tempo	SELECT on TRACKS · GLO + key 16 (tap) · GLO → GLOBAL → BPM
swing	TRACKS KNOB 1 (global) · SEQ + KNOB 3 (track) · GLO → GLOBAL
metronome	GLO → GLOBAL → CLICK
a track's level / pan	TRACKS KNOB 2 / KNOB 4 · GLO + KNOB 1–4
the drum kit	PRESETS on the drum track · DRUMS kit page KNOB 1
the drum level / reverb	DRUMS kit page KNOB 2 / 3 · GLO → DRUMS
the drum delay (2.5)	GLO → DRUMS → DLY (KNOB 4)
a track's filter	FX + KNOB 4 · FX → FILTER
delay time	FX → DLY → TIME
chords	SEL + KNOB 1 · SEL page

Looking for	It is
STRUM, VLEAD	SEL 2
your projects	SAVE + keys 1–4 (play) / 5–8 (save) · SAVE → PROJECT (LOAD, SAVE, ERASE)
your sounds	SAVE → USER · PRESETS (after the factory sounds)
MIDI out / clock / clock only	GLO → SYSTEM: MIDI · SYNC · IN
each track's MIDI channel	GLO → MIDI (CH1 CH2 CH3 CH4)
MIDI in / out, the other channels, the bend	GLO → MIDI 2 (IN OTHER OUT BEND)
a vinyl / tape / rain bed under the mix	GLO → TEXTURE (2.6)
MIDI LEARN	GLO + F#3 · forget: hold HOME → SYSTEM → MIDI LEARN CLEAR
a track's effect (INSERT)	FX → INSERT, INSERT 2
tempo-synced envelope / LFO	ENV DEST → ESYNC · LFO → LFO 2 → SYNC
ANALOG band-pass / high-pass	EDIT → FILTER → TYPE
a note's length in steps	SEQ → NOTES page, KNOB 3 on the note (LEN)
drum mutes	EDIT + a black key, then the sounds' keys
lights, colours, USB audio level	hold HOME (menu)
the version	hold HOME → SYSTEM → ABOUT

SLOOP is based on Felucca by Leo Kuroshita (@kurogedelic), Hügelton Instruments. Full credits and licences: SLOOP.md and LICENSING.md. M-VAVE and FM-1 are trademarks of their owners; SLOOP is not affiliated with them.